

# What has to be fixed in `dhkm.tex`

A complete problem report, ordered for formalization in Rocq

Checked against `dhkm.tex` (1163 lines, 2026-08-27), `coq/src/` (13 files), `README.md`,  
`tex/sections/biblio.bib`, and `WiL Abstract/Fuzzifying Derivatives of Regular  
Expressions.tex`

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## How to read this

Every problem below is numbered once, from 1 to 77, and the numbering is the suggested working order: fixing them top to bottom never makes you redo earlier work.

Each item says, briefly, what is wrong; then a counterexample where there is one; then a suggested fix. Nothing is explained twice.

Prose proofs in the paper are deliberately not reviewed. Every result is expected to carry a Rocq citation in the end, so an item appears below only when the *statement* is wrong, ambiguous, or not yet known to be true — never merely because the paper’s own argument is thin or missing.

The parts are ordered by what blocks what.

- **Part A** — the four decisions that change *what the theorems say*, plus the pending Rocq port they all feed into. Nothing after this part is worth polishing first. Item 5 is the one nobody has recorded before, and it forces a redesign.
- **Part B** — definitions Rocq cannot accept: ill-founded, infinite, ambiguous, or referring to something undefined. This is the hard gate.
- **Part C** — statements that are false, or not yet known to be true and load-bearing.
- **Part D** — smaller errors and editorial defects. Do these last, in one pass; some will be deleted outright by Part A.
- **Part E** — problems in the existing `coq/` development.
- **Part F** — what cannot be formalized at all yet, and why.
- **Part G** — the phase-by-phase plan for formalizing the whole paper.
- **Part H** — what I checked and found *correct*. Do not spend time there.

Tags: **[DECISION]** needs an author decision; **[BLOCKS ROCQ]** makes Rocq transcription impossible; **[FALSE]** is a false statement; **[UNPROVED]** is true-or-unknown and not machine-checked; **[ROCQ CODE]** is something to change in `coq/`; **[NOT YET DONE]** is work that is simply outstanding rather than a defect; **[EDITORIAL]** is cosmetic.

Line numbers refer to the current `dhkm.tex`. Result numbers (Definition 3.5, Theorem 4.2, ...) are the ones the compiled PDF prints — they were read off `dhkm.aux`, so they are what a reader sees. They have *shifted* since the earlier review notes: what `todo-list.md` calls Definition 3.6 is now **Definition 3.5**, and its Definition 3.7 is now **Definition 3.6**.

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## The three facts that shape everything else

1. **Theorem 4.2 is what the Rocq development has to deliver, and not one of its five items is yet in a state where a proof can be attempted.** Two of them rest on definitions that do not denote anything (Part B); one is stated only as an inclusion, with its true counterpart commented out; and two are not known to hold for a general norm. There is no existing argument to lean on either: the drafts in the 295 dead lines after `\end{document}` each assume idempotence silently or establish something false.
2. **The derivative is not a congruence for the paper’s  $\sim$  (item 5).** The paper’s automaton is therefore not well defined, and this is why the Rocq automata files look the way they do.
3. **The Rocq fuzzy layer still targets the previous version of the paper.** `coq/src/SimilarityRelations.v` is written for the minimum T-norm, so what it formalizes is the Gödel-only WiL development rather than `dhkm.tex`. That is a port waiting to happen once Section 3 is final, not an error — but `smn` occurs in the *statements* and not only in the proofs, so the port restates every fuzzy signature (item 1). Separately, the development does not currently compile at all (item 64).

## Part A — Settle these first

Item 1 is not a defect; it is the porting job that the whole plan in Part G feeds into, and it comes first because everything about the fuzzy layer is downstream of it. Items 2–5 are genuine decisions: each one changes what the theorems say.

1. **The fuzzy Rocq layer is written for the minimum T-norm and still has to be ported to a general one** [NOT YET DONE] [ROCQ CODE]

Where: Sections 3–4 of `dhkm.tex` vs. `coq/src/SimilarityRelations.v`:26, 143, 149, 725, 1125

**Situation.** Nothing here is wrong. `SimilarityRelations.v` was written for the earlier, Gödel-only version of the development, so it defines `smn` as *the* T-norm; `dhkm.tex` now quantifies over an arbitrary left-continuous  $\otimes$ . The port is simply outstanding, and it should stay outstanding until Section 3’s statements are final — which is what the rest of this report is about. It is listed first only because its size is what makes the ordering in Part G matter.

**Note.** What the port will cost, so that it can be scheduled rather than discovered. `smn` occurs in the *statements* and not only in the proofs: inside `is_similarity`, inside the hypothesis `R_trans`, and in the definitions of `Rw`, `Rr` and `RL`. So every fuzzy signature is restated, not merely reproved. The proof *structures* do survive, and the mechanical part of the work is deciding, at each `smn`, whether it is “genuinely  $\otimes$ ” or “genuinely min” — note that  $\mathcal{R}^E$ ’s  $+$  clause and  $\mathcal{R}^L$ ’s outer combination really are min, not  $\otimes$ , so a blind search-and-replace would be wrong.

**Fix.** Build the T-norm interface of Phase 3 and instantiate it at min first: the existing files then port almost unchanged and serve as a regression test for the interface. Generalize afterwards. See Phases 3 and 4 in Part G.

2. **The paper claims to generalize a WiL theorem that it elsewhere refutes** [DECISION] [FALSE]

Where: line 534 (“This section is a generalization of the results in [KM26]”); WiL abstract, Theorem “Semantic–syntactic correspondence”

**Problem.** The WiL abstract states  $\mathcal{L}(\mathcal{D}_a^\mu(r)) = \partial_a^\mu(\mathcal{L}(r))$ . That equality is false — under WiL’s own Gödel norm. `dhkm.tex` already contains the refutation (Remark 4.2) and correctly weakens

the live version to  $\subseteq$  (Theorem 4.2(3)). So the paper refutes and claims to generalize the same statement, 250 lines apart.

**Counterexample.**  $\Sigma = \{a, b\}$ ,  $\otimes = \min$ ,  $\mathcal{R}(a, b) = \mu < 1$ ,  $r = a^*$ . Then  $\mathcal{R}^E(b^*, a^*) = \mu$ , so  $b^*$  is in the index set and  $\mathcal{D}_a^\mu(a^*) \sim a^* + b^*$ , giving  $\mathcal{L}(\mathcal{D}_a^\mu(a^*)) = \{a\}^* \cup \{b\}^*$ . But  $ab \in \partial_a^\mu(\{a\}^*)$ : take  $b' = a$  and  $v = aa$ , so that  $\mathcal{R}(a, a) = 1$ ,  $\mathcal{R}^w(ab, aa) = \min\{1, \mathcal{R}(b, a)\} = \mu$  and  $aaa \in \{a\}^*$ . And  $ab \notin \{a\}^* \cup \{b\}^*$ , so  $\supseteq$  fails.

**Fix.** Rewrite line 534 to say which WiL results are *generalized* and which are *corrected*, and state the correction (item 19). Separately, WiL iterates  $\partial^\mu$  by plain composition with base case  $\partial_\varepsilon^\mu(L) = L$ , while `dhkm.tex` Definition 4.6 uses a union with base case  $L^\mu$ : say which is intended and prove they agree under  $\mu \otimes \mu = \mu$  (item 36).

### 3. 295 lines — including the whole Automata section — sit after `\end{document}` [DECISION]

Where: `\end{document}` at line 868; dead text 870–1163

**Problem.** The compiled PDF has no Automata section, yet the abstract (line 40) and the introduction (line 52) both promise that derivatives “give a direct construction of automata from regular expressions”, and `README.md` lists `Automata.v` and `dfa_correctness` as formalized. What is dead is five *statements*: fuzzy regularity, the fuzzy correspondence, both fuzzy finiteness theorems, the crisp derivative DFA, and the fuzzy DFA. The last two appear nowhere in the live text, and they are the two the Rocq development is actually aimed at — a result cannot cite a Rocq proof if the result is not in the document.

**Fix.** Decide now; it is a scope decision, not an edit. Reviving is the only choice consistent with the abstract. The prose arguments in the block are not what is worth saving — each of them assumes idempotence silently or establishes something false (items 5, 14, 18–20) — so revive the statements, in the corrected form those items describe, and let the proofs be Rocq’s. If you drop the block instead, delete the automata promise from the abstract and the introduction.

### 4. Two different relations are both called $\sim$ , and the paper’s Lemma 2.1 is a tautology in Rocq [DECISION] [ROCQ CODE]

Where: Definition 2.2 and Lemma 2.1, lines 104–163, vs. `Canonicalization.v:215` and `CanonicalizationCorrectness.v:673`

**Problem.** The paper defines  $\sim$  as the relation “generated by” a list of equations, then states as a *lemma* that  $r \sim s \iff \text{nf}(r) = \text{nf}(s)$ . Rocq *defines* `similar r s := eq_regex (nf r) (nf s)`, so `similar_iff_nf_eq` merely reflects boolean structural equality into propositional equality — a two-line tautology. There is no inductively defined `similar` anywhere in `coq/src/`. So the paper’s  $\sim$  and the formalized  $\sim$  are different relations, the real content of Lemma 2.1 (completeness of the normalizer with respect to the equational theory) is not formalized, and Lemma 2.1’s proof points at that tautology — so the citation does not support the lemma it is attached to. Since every result is meant to be backed by a Rocq citation, a pointer that resolves to a tautology is worse than none.

**Fix.** Pick one reading — it decides how much is left to prove. (i) Keep  $\sim$  equational, state Lemma 2.1 honestly as soundness *and* completeness, add an inductive `similar` to Rocq, and mark completeness unformalized. This matches Coquand–Siles. (ii) Define  $\sim$  in the paper as `nf`-equality. Then Lemma 2.1 is a definition, Lemma 2.2 carries all the content, and Definition 2.2’s equations become a description of what `canonize` implements. This matches the code and is far cheaper. Under either reading, say  $\sim$  is the least **congruence** containing the equations: read literally, “generated by” gives only the closure of the axiom *instances*, so  $(r + r) \cdot s \sim r \cdot s$  is not derivable — yet `canonize` rewrites under contexts (`Canonicalization.v:186–197`).

**5. The derivative is *not* a congruence for Definition 2.2's  $\sim$ , so the paper's automaton is not well defined** [DECISION] [FALSE] [BLOCKS ROCQ]

Where: Definition 2.2 (line 104); the DFA definition in the dead block (line 1056); `Canonicalization.v`

**Problem.** Brzozowski's own similarity is associativity, commutativity and idempotence of  $+$  only, and  $\mathcal{D}_a$  respects it. Definition 2.2 adds  $\cdot$ -associativity, the unit and annihilator laws,  $\emptyset^* \sim \varepsilon$ ,  $\varepsilon^* \sim \varepsilon$ ,  $(r^*)^* \sim r^*$ , &-ACI and  $\neg\neg$ .  $\mathcal{D}_a$  **does not respect those**. The dead block's DFA sets  $Q = \{[\mathcal{D}_u(r)]\}$  and  $\delta(q, a) = [\mathcal{D}_a(q)]$ ; for that to be well defined one needs  $[\mathcal{D}_a([t])] = [\mathcal{D}_a(t)]$ , which fails. So  $\delta$  can leave  $Q$ , and  $[\mathcal{D}_a([\mathcal{D}_u(r)])] \neq [\mathcal{D}_{ua}(r)]$ .

**Counterexample.** Both computed with `Canonicalization.v`'s own `canonize`. (a)  $(a^*)^* \sim a^*$ , since both normalize to  $a^*$ . But  $\mathcal{D}_a((a^*)^*) = (\varepsilon \cdot a^*) \cdot (a^*)^*$  normalizes to  $a^* \cdot a^*$ , while  $\mathcal{D}_a(a^*) = \varepsilon \cdot a^*$  normalizes to  $a^*$  — and  $a^* \cdot a^* \not\sim a^*$ , because the law  $r^* \cdot r^* \sim r^*$  is not in the list. (b) Associativity fails too:  $(a^* \cdot a) \cdot a \sim a^* \cdot (a \cdot a)$  (both normalize to  $a^* \cdot (a \cdot a)$ ), but  $\mathcal{D}_a$  of the first normalizes to  $(\varepsilon + a^* \cdot a) \cdot a$  and  $\mathcal{D}_a$  of the second to  $a + a^* \cdot (a \cdot a)$  — different top-level constructor, so certainly not  $\sim$ .

**Fix.** Three options. (i) Use ACI-of- $+$  only for the automaton, as Brzozowski does. Then  $\mathcal{D}_a$  is a congruence,  $\delta$  is well defined,  $Q = \{[\mathcal{D}_u(r)]\}$  is correct, and Brzozowski's finiteness theorem applies verbatim. The extra Coquand–Siles laws stay as an optional state-reducing simplification applied *outside* the automaton definition. (ii) Keep the extended  $\sim$  but define  $Q$  as the set reachable from  $[r]$  under  $q \mapsto [\mathcal{D}_a(q)]$ , not as  $\{[\mathcal{D}_u(r)]\}$  — then prove *that* set finite. This is a new obligation, it is not obviously true, and it is exactly why `AutomataConstruction.v` needs a `fuel` parameter (item 72). (iii) Add laws until  $\mathcal{D}_a$  becomes a congruence. Do not: by Redko there is no finite equational system to converge to. Recommended: (i) for the definition in the paper, (ii) for the executable construction.

**Note.** Theorem 4.1(4) (“finite up to canonicalization”) survives either way, because `nf`-classes are coarser than ACI-classes. It is only  $\delta$ 's well-definedness that breaks.

## Part B — Definitions Rocq cannot accept as written

### 6. Definition 4.7 sums over a continuum [BLOCKS ROCQ]

Where: lines 483–491

**Problem.**  $\mathcal{D}_{av}^\mu(r) = \sum_{\mu' \otimes \mu'' \geq \mu} \mathcal{D}_v^{\mu'}(\mathcal{D}_a^{\mu''}(r))$  indexes over all pairs  $(\mu', \mu'') \in [0, 1]^2$  with  $\mu' \otimes \mu'' \geq \mu$  — a continuum — and combines them with regular-expression  $+$ , which is finitary. A regular expression is a finite syntactic object, so this is not a definition. **Nothing in Section 4.2 is transcribable until this is fixed**, because Theorem 4.2’s items 1, 4 and 5 all mention  $\mathcal{D}_u^\mu$ .

**Fix.** The summands take only finitely many *distinct* values: as  $\mu''$  decreases, the index set  $\{(b, s) \mid \mathcal{R}(a, b) \otimes \mathcal{R}^E(s, r) \geq \mu''\}$  runs through a finite increasing chain of subsets of a finite set (item 25). So restrict  $(\mu', \mu'')$  to the finite set of values actually taken by  $\mathcal{R}(a, \cdot) \otimes \mathcal{R}^E(\cdot, r)$ , or define the sum over the finite set of distinct summands — and extend the note at line 491 to cover it. As written, that note addresses Definition 4.5’s sum over  $(b, s)$ , not this  $\mu'$ -continuum.

**Note.** Do *not* collapse it to plain iteration  $\mathcal{D}_v^\mu(\mathcal{D}_a^\mu(r))$ : plain iteration accumulates tolerance  $\mu \otimes \mu$ , not  $\mu$ , so it is strictly smaller unless  $\mu \otimes \mu = \mu$ . The semantic side (Definition 4.6) keeps its union for the same reason, so the two sides currently agree by construction.

### 7. $L^\mu$ is used for a language but defined only for an expression, and $\sim_\mu^{\Sigma^*}$ is never defined [BLOCKS ROCQ]

Where: Definition 3.6 (line 279 — which also has no `\label`); Definition 4.6’s base case (line 476)

**Problem.** Definition 3.6 reads  $\mathcal{L}^\mu(r) = \{w \mid w \sim_\mu^{\Sigma^*} v \text{ for } v \in \mathcal{L}(r)\}$ . Three defects. (a)  $\sim_\mu^{\Sigma^*}$  appears nowhere in Section 3; it is introduced only in Proposition 4.1(3), in Section 4.3, under a hypothesis Section 3 does not assume. (b) “for  $v \in \mathcal{L}(r)$ ” reads universally, which gives the wrong — usually empty — set; it must be “for *some*  $v$ ”. (c) Definition 4.6’s base case  $\partial_\varepsilon^\mu(L) = L^\mu$  applies  $L^\mu$  to a *language*, which is not defined anywhere.

**Fix.** In Section 3, unconditionally, define the  $\mu$ -cut of  $\mathcal{R}^w$ , and then for any  $L \subseteq \Sigma^*$  put  $L^\mu = \{w \in \Sigma^* \mid \exists v \in L. \mathcal{R}^w(w, v) \geq \mu\}$ ; Definition 3.6 becomes the instance  $\mathcal{L}^\mu(r) := (\mathcal{L}(r))^\mu$ . Do the same for  $\sim_\mu^E$ , which is used at line 469 and defined only at line 545. Add a `\label` to Definition 3.6.

**Note.**  $\mathcal{L}^\mu$  is the fuzzy semantics on which Theorem 4.2(1), Theorem 3.1’s “Moreover” clause and the corrected correspondence all rest. It is the most load-bearing undefined symbol in the paper.

### 8. Is $\mathcal{R}^E$ defined on restricted expressions or on all of $E$ ? The paper says both [BLOCKS ROCQ] [ROCQ CODE]

Where: Definition 3.5 (line 255) vs. Lemma 3.2(1) (line 272); `SimilarityRelations.v:1125`

**Problem.** Definition 3.5 says “let  $r, s$  be restricted” and has no clauses for  $\&$  or  $\neg$ . Lemma 3.2(1) claims  $\mathcal{R}^E$  is a T-similarity “on  $E$  and hence on restricted regular expressions”. If the final clause “0 otherwise” is read as total, then  $\mathcal{R}^E$  on  $E$  does exist but is the identity off the fragment — which is exactly why the  $(\&)$  counterexample later misfires (item 22). These are different Rocq constants and different theorems, and the choice propagates into every fuzzy signature.

**Fix.** State Definition 3.5’s domain once, and make Lemma 3.2(1) match. Recommended: restricted only.

**Note.** The code has already made the opposite choice: `Rr` is defined on all of `regex`, with `Rr` (And `r1 r2`) (And `s1 s2`) = `smin ...` and `Rr` (Neg `r1`) (Neg `s1`) = `Rr r1 s1` — precisely the extension the paper’s counterexamples exist to refute. One of the two has to move.

**9.  $\mathcal{G}^E$  is a relation, not a function: the skeleton  $r[a_1, \dots, a_n]$  is not unique** [BLOCKS ROCQ]

Where: notation paragraph, line 97; Definition 3.4 (line 237)

**Problem.** The notation paragraph says  $r[a_1, \dots, a_n]$  highlights *some* occurrences of alphabet symbols. The same pair  $(r_1, r_2)$  therefore admits many decompositions, and Definition 3.4 does not say which to use — so as written  $\mathcal{G}^E$  is a relation, not a function. It happens to be well defined *for min*: an unhighlighted position must carry the same symbol in both expressions and so contributes  $\mathcal{R}(a, a) = 1$ , which never changes a min. But that is an accident of the Gödel norm, and it is exactly what fails for a general  $\otimes$  — which is part of why Definition 3.5 has to be inductive.

**Fix.** Require the skeleton in which *every* symbol occurrence is highlighted. Definition 4.8(2) (line 618) already says “all the alphabet symbols occurring in  $r$ ” — hoist that convention to line 97 and to Definition 3.4, and the well-definedness gap closes everywhere at once.

**10.  $\mathcal{G}^E$  and  $\mathcal{R}^E$  are used interchangeably in the very section that argues they differ** [BLOCKS ROCQ]

Where:  $\mathcal{R}^E$  written for  $\mathcal{G}^E$  at lines 253, 306, 311, 327, 330, 333

**Problem.** These are different relations off the Gödel norm, and that difference is the entire point of Section 3. Each site is a different Rocq constant and the statements differ accordingly.  $\mathcal{R}^E$  is even used at line 253, before Definition 3.5 exists.

**Fix.** Use  $\mathcal{G}^E$  for Definition 3.4 throughout and  $\mathcal{R}^E$  only after Definition 3.5. Add the remark that Definition 3.5 restricts to Definition 3.4 when  $\otimes = \min$  — it does; I checked all five clauses.

**11. A sum over a finite index set is not yet a syntactic object** [BLOCKS ROCQ]

Where: Definitions 4.5 (line 457) and 4.7 (line 483)

**Problem.**  $+$  is binary. A sum over a *set* of pairs is well defined only up to associativity, commutativity and idempotence of  $+$ . To be a function into `regex` — which is what Rocq needs — it requires a fixed enumeration order and duplicate removal. This is invisible on paper and fatal in Rocq.

**Fix.** Either say the index set is enumerated in a fixed order (for instance, the `nf` of the right-nested sum over a sorted, duplicate-free list), or state explicitly that  $\mathcal{D}_a^\mu(r)$  is defined only up to  $\sim$  and that every downstream statement is  $\sim$ -invariant. In Rocq: build the index set as a sorted `seq` (`A * regex`) and fold with `Alt` (Phase 5).

**12.  $\otimes^n$ ,  $\inf_n$ , the empty  $\otimes$ -product,  $\inf \emptyset$  and  $\sup \emptyset$  are load-bearing and never stated** [BLOCKS ROCQ]

Where: Definition 3.2 (line 220), Definition 3.3 (line 225), Definition 3.5’s star clause (line 261)

**Problem.** Four omissions, each of which changes a value. (a) In Definition 3.2 with  $n = 0$  the clause is an empty  $\otimes$ -product; the convention  $= 1$ , giving  $\mathcal{R}^w(\varepsilon, \varepsilon) = 1$ , is what makes  $\mathcal{R}^w$  reflexive. (b) Definition 3.3 takes  $\inf$  and  $\sup$  over possibly empty languages; the conventions  $\inf \emptyset = 1$  and  $\sup \emptyset = 0$  are needed *and used* — the (&) counterexample at line 329 relies on  $\mathcal{R}^L(\{a\}, \emptyset) = 0$ , which holds under exactly these conventions, and  $\mathcal{R}^L(\emptyset, \emptyset) = 1$  is what reflexivity requires. (c)  $\otimes^n x$ , the  $n$ -fold  $\otimes$ -power, is never defined. (d)  $\inf_n$  does not say whether  $n$  starts at 0 or 1.

**Fix.** State all four, in Section 3 or the appendix. The code already picks the same ones — `SupL_empty = 0` and `InfL_empty = 1`, `SimilarityRelations.v:394` and `406` — so this is purely a paper-side omission.

### 13. Definition 4.5’s index set has $b$ and $s$ free [BLOCKS ROCQ]

Where: lines 459–465

**Problem.**  $\mathcal{D}_a^\mu(r) = \sum_{\mathcal{R}(a,b) \otimes \mathcal{R}^E(s,r) \geq \mu} \mathcal{D}_b(s)$  never says what  $b$  and  $s$  range over.

**Fix.** “the sum ranges over all  $b \in \Sigma$  and all restricted regular expressions  $s$  such that...”. Also record explicitly that  $(b, s) = (a, r)$  always qualifies, since  $\mathcal{R}(a, a) \otimes \mathcal{R}^E(r, r) = 1$ : the index set is therefore never empty, and  $\mathcal{D}_a(r)$  is always a summand — which is what gives  $\mathcal{L}(\mathcal{D}_a(r)) \subseteq \mathcal{L}(\mathcal{D}_a^\mu(r))$  for free.

### 14. $[\cdot]$ means four things, and $\mathcal{L}$ means two [BLOCKS ROCQ]

Where: Lemma 2.1 ( $[r] = \text{nf}(r)$ ); Definition 4.8 ( $[a]_\mu$ ,  $[L]_\mu$ , and  $[r]_\mu$  at line 693);  $\mathcal{L}$  as a bound language variable at lines 473 and 583

**Problem.** Section 4.3 needs canonicalization *and* quotients in the same breath, so at least one bracket must change symbol. Worse, Definition 4.6 and Proposition 4.3 use  $\mathcal{L}$  as a bound *language variable* while  $\mathcal{L}(r)$  is the semantics map. Two consequences are already visible: the proof of Proposition 4.4 (lines 660–686) writes  $\mathcal{L}^\mu$  throughout for the quotient image  $[L]_\mu$  — a different set over a different alphabet — and the “Hence” display at line 650 writes  $[\partial_a^\mu(L)]^\mu$  with a superscript where the definition uses a subscript.

**Fix.** Keep  $[r]$  for nf; write  $q_\mu(a)$ ,  $q_\mu^*(w)$ ,  $q_\mu^*(r)$  for quotients (the paper already has those maps); use plain  $L$  for language variables. Then fix the Proposition 4.4 proof, which is currently flagged “not checked, the notation has changed also”.

### 15. $\nu$ ’s displayed equation emits $\&$ , which leaves the restricted fragment [BLOCKS ROCQ]

Where: Definition 4.1, line 366

**Problem.**  $\nu$  is *defined* semantically (“returns  $\varepsilon$  if  $r$  is nullable and  $\emptyset$  otherwise”), so its output is always  $\varepsilon$  or  $\emptyset$ , and  $\mathcal{D}_a$  really does preserve restrictedness. But the displayed characterization writes  $\nu(r \cdot s) = \nu(r) \& \nu(s)$ , putting  $\&$  into a fragment that has none — and Theorem 4.2’s restricted-expression claims then look unjustified. Since  $\mathcal{R}^E$  is defined only on restricted expressions and Definition 4.5 sums  $\mathcal{D}_b(s)$  over restricted  $s$ , this matters.

**Fix.** Write  $\nu(r \cdot s) = \nu(r) \cdot \nu(s)$ , which is equally correct ( $\varepsilon \cdot \varepsilon \sim \varepsilon$ , and  $\emptyset$  in either argument gives  $\emptyset$ ), and the closure lemma is free. `coq/src/Nullable.v:30` already does the right thing.

**Note.** The  $\nu$  equations are identities only *up to*  $\sim$ :  $\nu(a + a) = \emptyset + \emptyset$ , not  $\emptyset$ . Say “up to  $\sim$ ” or compose with nf.

## Part C — False, or not yet known, and load-bearing

**16. Theorem 4.2, the Main Theorem, is not yet in a state where a Rocq proof can be attempted [UNPROVED]**

Where: lines 498–524

**Problem.** This is the theorem the whole formalization is aimed at, and not one of its five items can be transcribed today. Items 1 and 4 rest on Definition 4.7, which does not denote anything (item 6), and item 1 also rests on  $\mathcal{L}^\mu$ , which is undefined for the case it needs (item 7); item 3 is stated only as an inclusion, with its true counterpart commented out (item 19); and items 2, 4 and 5 are not known to hold for a general norm. Nor is there an existing argument to lean on: the drafts in the dead block each assume idempotence silently or establish something false (items 3, 18–20).

**Fix.** Items 17–20 say what each of the five statements needs; Phases 5–7 in Part G say in what order to attack them.

**17. Theorem 4.2(1), the *experimentum crucis*: right in kind, but not yet a statement about anything [UNPROVED]**

Where: lines 500–501

**Problem.** Four defects, none of which is about truth. (a) It rests on Definition 4.7, a continuum sum (item 6), and on  $\mathcal{L}^\mu$ , which is undefined for the case it needs (item 7). (b) Wrong domain: it says “for every regular expression  $r$ ”, but  $\mathcal{D}^\mu$  and  $\mathcal{R}^E$  are defined only on *restricted* expressions (item 8). (c) “If  $\Sigma$  is finite” is redundant — Definition 2.1 already makes  $\Sigma$  finite — and the sentence “If  $\Sigma$  is finite for every regular expression  $r$ ” is ungrammatical. (d) It is stated extensionally, but what makes it *crucial* is that it yields a decision procedure:  $\mathcal{D}_a^\mu(r)$  is computable once item 6 is fixed, and nullability is decidable, so fuzzy membership is decidable. Say that.

**Note.** I could not break it. Verified on: Gödel min with  $\Sigma = \{a, b\}$ ,  $\mathcal{R}(a, b) = \mu = 0.5$ ,  $r \in \{a^*, a \cdot a\}$ ,  $u \in \{ab, ba, bb\}$ ; and Goguen product with  $\mu \in \{0.25, 0.5\}$ ,  $r = a \cdot a$ ,  $u \in \{ab, ba, bb\}$ . It holds in every case, including on the Gödel example that refutes the WiL correspondence.

Worth adding to the paper, because it is currently stated nowhere: item 1 is an *equality* while item 3 is only an *inclusion* precisely because item 1 compares total tolerance on both sides, whereas items 3 and 4 compare per-step objects — which is why those need  $\mu \otimes \mu = \mu$  to line up. That contrast is the paper’s best insight into its own machinery and it pre-empts items 19, 21 and 24 at once.

**Note.** The decision-procedure reading in (d) is only achievable if  $\mathcal{R}^E(s, r) \geq \mu$  is decidable, which Definition 3.5’s star clause makes false for a general norm. See F2.

**18. Theorem 4.2(2), regularity: not known for a general norm, and much harder than it looks [UNPROVED]**

Where: line 503; draft argument at line 882 (dead)

**Problem.** The second half (“if  $r$  is a regular expression then  $\mathcal{D}_a^\mu(r)$  is a regular expression”) is trivial and should say *restricted*. The first half —  $\partial_a^\mu(L)$  regular for regular  $L$  — is the content, and it is not known for a general  $\otimes$ . The one existing argument establishes something weaker: it routes through the quotient homomorphism of Proposition 4.4, which holds only under  $\mu \otimes \mu = \mu$ , and it uses plain iteration for  $\partial_a^\mu$ , which is not what Definition 4.6 says.

**Note.** Why it is genuinely hard for a general norm: the membership test is  $\mathcal{R}(a, b) \otimes \otimes_i \mathcal{R}(w_i, v_i) \geq \mu$ , a *threshold on a  $\otimes$ -product along the word*, not a letterwise condition. So  $\partial_a^\mu(L)$  is a cut-point

language of a  $\max \otimes$  weighted automaton. It *is* provable for continuous norms by a bounded-budget argument — factors that are  $\otimes$ -idempotent are free and reduce to a letterwise condition, while within each Archimedean piece of the ordinal-sum decomposition only boundedly many non-idempotent factors can occur before the product falls below  $\mu$  — but that is a substantial theorem in its own right, and nobody has written it down.

**Fix.** Either commit to the bounded-budget argument, or state item 2 under  $\mu \otimes \mu = \mu$ , where the quotient proof is correct, and flag the general case as open.

**19. Theorem 4.2(3), the weak correspondence, is true, but the reverse half is missing [UNPROVED]**

Where: line 505; the commented-out item 4 at line 509

**Problem.** Item 3,  $\mathcal{L}(\mathcal{D}_a^\mu(r)) \subseteq \partial_a^\mu(\mathcal{L}(r))$ , holds for every left-continuous  $\otimes$  — I checked the proof: it uses  $\mathcal{R}(a, b') \geq \mathcal{R}(a, b) \otimes \mathcal{R}(b, b')$  and Theorem 3.1(1), with the supremum attained because  $\Sigma$  is finite and  $\mathcal{R}^w$  vanishes off equal lengths. Commenting out item 4 removed a falsehood, but it also removed two things that were not wrong: the reverse inclusion, and the equality that *is* true under  $\mu \otimes \mu = \mu$ .

**Fix.** Restore item 4 as the two-sided inclusion, which holds for every left-continuous  $\otimes$ :

$$\mathcal{L}(\mathcal{D}_a^\mu(r)) \subseteq \partial_a^\mu(\mathcal{L}(r)) \subseteq \mathcal{L}^\mu(\mathcal{D}_a^\mu(r)).$$

The right-hand inclusion is three lines:  $w \in \partial_a^\mu(\mathcal{L}(r))$  gives  $b', v$  with  $\mathcal{R}(a, b') \otimes \mathcal{R}^w(w, v) \geq \mu$ ; since  $\otimes \leq \min$ , both factors are separately  $\geq \mu$ , so  $(b', r)$  is in the index set of  $\mathcal{D}_a^\mu(r)$  and  $v \in \mathcal{L}(\mathcal{D}_{b'}(r))$ , hence  $w \in \mathcal{L}^\mu(\mathcal{D}_a^\mu(r))$ . Then add, as a *separate* result carrying  $\mu \otimes \mu = \mu$  in its own statement:  $\partial_a^\mu(\mathcal{L}(r)) = \mathcal{L}^\mu(\mathcal{D}_a^\mu(r))$ . I checked this: compose item 3 with  $\mathcal{R}^w(w, v) \otimes \mathcal{R}^w(v, v') \geq \mu \otimes \mu = \mu$ . Note that the *other* equality,  $\mathcal{L}(\mathcal{D}_a^\mu(r)) = \partial_a^\mu(\mathcal{L}(r))$ , is false even for Gödel — that is Remark 4.2.

**Counterexample.** Separating the two inclusions (verified numerically).  $\Sigma = \{a, b, c\}$ , Goguen product,  $\mu = 0.5$ ,  $\mathcal{R}(x, x) = 1$ ,  $\mathcal{R}(a, b) = \mathcal{R}(b, c) = 0.5$ ,  $\mathcal{R}(a, c) = 0.25$ . This is a genuine extensional T-similarity for the product norm —  $\mathcal{R}(a, c) = \mathcal{R}(a, b) \cdot \mathcal{R}(b, c)$  is tight and all 27 triples check. Take  $r = b \cdot b$ , so  $\mathcal{L}(r) = \{bb\}$ . Then  $\partial_a^\mu(\mathcal{L}(r)) = \{b\}$ ; the index set of  $\mathcal{D}_a^\mu(r)$  is  $\{(a, bb), (a, ab), (a, ba), (a, cb), (a, bc), (b, bb)\}$  and only  $\mathcal{D}_a(a \cdot b)$  and  $\mathcal{D}_b(b \cdot b)$  survive, so  $\mathcal{D}_a^\mu(r) \sim b$  and  $\mathcal{L}(\mathcal{D}_a^\mu(r)) = \{b\}$ ; but  $\mathcal{L}^\mu(\mathcal{D}_a^\mu(r)) = \{a, b, c\}$ , since  $\mathcal{R}(a, b) = \mathcal{R}(c, b) = 0.5 \geq \mu$ . So the second inclusion is strict, and the equality needs the hypothesis. Keep this counterexample — it is what justifies the local hypothesis.

**20. Theorem 4.2(4) and (5), finiteness: type errors, double binding, and not known for a general norm [UNPROVED]**

Where: lines 507–513; draft arguments at lines 978 and 1013 (dead)

**Problem.** (a) Item 4 writes  $\mathcal{D}_a^\mu(L)$ , applying the *syntactic* derivative to a language; it should be  $\mathcal{D}_a^\mu(r)$ . (b) Both items bind  $u$  twice: “for every word  $u \in \Sigma^*$  ... the set  $\{\partial_u^\mu(L) \mid u \in \Sigma^*\}$ ”. Drop the outer quantifier. (c) Both existing arguments use the quotient ( $\mu \otimes \mu = \mu$ ) and plain iteration, neither of which is available for a general norm. Whether the two results survive without idempotence is, as far as I can tell, open.

**Fix.** Fix the types, then either settle them generally or state them under  $\mu \otimes \mu = \mu$ . If they need the hypothesis, so does the automata construction that depends on them — say so explicitly, because that is the one place the hypothesis silently propagates furthest.

**21. Remark 4.2’s displayed derivative silently assumes idempotence, and has a type error [FALSE]**

Where: line 526

**Problem.** It asserts  $\mathcal{D}_a^\mu(a^*) = a^* + b^*$  given  $\mathcal{R}(a, b) = \mu < 1$ . That needs  $\mathcal{R}^E(b^*, a^*) \geq \mu$ , and by Definition 3.5  $\mathcal{R}^E(b^*, a^*) = \inf_n \otimes^n \mu$ , which equals  $\mu$  only when  $\otimes$  is idempotent at  $\mu$ . For the product norm it is 0, and then  $\mathcal{D}_a^\mu(a^*) \sim a^*$ .

**Note.** The remark still refutes item 3 either way —  $ab \in \partial_a^\mu(\{a\}^*)$  because  $1 \otimes \mathcal{R}^w(ab, aa) = \mathcal{R}(b, a) = \mu \geq \mu$ , while  $ab \notin \mathcal{L}(a^*)$  — so only the displayed value needs the proviso.

**Fix.** Add the proviso. Also: “ $ab \notin (a^* + b^*)$ ” must be “ $ab \notin \mathcal{L}(a^* + b^*)$ ”;  $r$  is never bound in “ $\partial_a^\mu(\mathcal{L}(r))$ ”; and the remark does not say which item it refutes (item 3, now the theorem’s only inclusion).

**22. The (&) counterexample contradicts Definition 3.5, and the clause that actually needs a counterexample has none [FALSE]**

Where: lines 324–336

**Problem.** (a) The (&) item claims  $\mathcal{R}^E(a \& a, a \& b) = \mathcal{R}(a, b)$ . Under Definition 3.5 that value is 0: there is no & clause, so “0 otherwise” applies. The claim is true of  $\mathcal{G}^E$  (Definition 3.4), whose skeleton  $x \& y$  does match, giving  $\min\{1, \mathcal{R}(a, b)\}$ . The counterexample is aimed at the wrong definition, and the alleged failure of Theorem 3.1 does not occur. (b) The ( $\neg$ ) item’s arithmetic is correct — I checked both  $\mathcal{R}^L(\mathcal{L}(\neg(a + a)), \mathcal{L}(\neg(a + b))) = 0$  for  $\Sigma = \{a, b\}$  and the general formula  $\mathcal{R}^L(\mathcal{L}(\neg a), \mathcal{L}(\neg b)) = \min\{\sup_{c \neq a} \mathcal{R}(a, c), \sup_{c \neq b} \mathcal{R}(b, c)\}$  — but it computes  $\mathcal{R}^E$  on the arguments of  $\neg$ , so it too presupposes a  $\neg$  clause the paper does not have. (c) The framing at line 324, “Theorem 3.1 fails for regular expressions when & and  $\neg$  are allowed”, is not well posed: Theorem 3.1 is stated only for restricted expressions and  $\mathcal{R}^E$  is defined only for them. The passage is really arguing that *no* extension can work, which is a different and stronger claim.

**Fix.** State the hypothetical clauses before the counterexamples: “were we to extend Definition 3.5 with  $\mathcal{R}^E(r_1 \& r_2, s_1 \& s_2) = \min\{\mathcal{R}^E(r_1, s_1), \mathcal{R}^E(r_2, s_2)\}$  and  $\mathcal{R}^E(\neg r, \neg s) = \mathcal{R}^E(r, s)$ , then Theorem 3.1(1) would fail:…”. Then both land. The  $\neg$  one is the stronger of the two: it survives any choice of clause, because the value depends on which *other* symbols are in  $\Sigma$ .

**23. Counterexample 2 for Definition 3.4 refutes a definition that is not in the paper [FALSE]**

Where: lines 251–253

**Problem.** The text says: assume  $\mathcal{R}(a, c) = \mathcal{R}(b, d) = 0.5$  and the Goguen product norm, then  $\mathcal{R}^L(\mathcal{L}(r_1), \mathcal{L}(r_2)) = 0.5$  while  $\mathcal{R}^E(a + b, c + d) = 0.5 \cdot 0.5 = 0.25$ . Verified numerically, all three of these values are 0.5:  $\mathcal{G}^E(a + b, c + d) = \min\{0.5, 0.5\} = 0.5$ ;  $\mathcal{R}^E(a + b, c + d) = 0.5$  because Definition 3.5’s  $+$  clause is min; and  $\mathcal{R}^L(\{a, b\}, \{c, d\}) = 0.5$ . There is no discrepancy to report. The 0.25 can only come from a hypothetical  $\otimes$ -everywhere extension of Definition 3.2 to expressions, which is never written down. Also  $r_1$  and  $r_2$  are never bound. Consequence: “too strict for  $+$ ” is false of Definition 3.4, which is min everywhere. What Definition 3.4 actually gets wrong besides  $*$  is **concatenation**, and in the same direction — too generous.

**Fix.** Replace item 2 with the concatenation counterexample: same  $\mathcal{R}$ , product norm,  $\mathcal{G}^E(a \cdot b, c \cdot d) = \min\{0.5, 0.5\} = 0.5$  but  $\mathcal{R}^L(\mathcal{L}(a \cdot b), \mathcal{L}(c \cdot d)) = \mathcal{R}(a, c) \cdot \mathcal{R}(b, d) = 0.25$ , violating Theorem 3.1(1). That is what motivates the  $\otimes$  clause for  $\cdot$  in Definition 3.5. Then restate the lead-in as two separate observations: the min-based Definition 3.4 is *too generous* for  $\cdot$  and for  $*$  (hence  $\otimes$  and  $\inf_n \otimes^n$  in Definition 3.5), and the naive  $\otimes$ -everywhere extension is *too strict* for  $+$  (0.25 against the true 0.5, hence min for  $+$ ).

**Note.** Counterexample 1 in the paper is correct as it stands:  $\mathcal{R}^L(\mathcal{L}(a^*), \mathcal{L}(b^*)) = \inf_n \otimes^n \mathcal{R}(a, b)$ , which equals  $\mathcal{R}(a, b)$  exactly when  $\mathcal{R}(a, b)$  is  $\otimes$ -idempotent.

#### 24. The red Proposition 4.2 is false, and the false item is now its only content [FALSE]

Where: lines 568–581

**Problem.** Items 1 and 2 are commented out, so the surviving item is the entire proposition. It claims that  $w_1 \sim_{\mu}^{\Sigma^*} w_2 \iff \mathcal{R}^w(w_1, w_2) \geq \mu^{|w_1|}$  is an equivalence, under the hypothesis “let  $\mathcal{R}^{\mu}$  be an equivalence” — strictly weaker than the  $\mu \otimes \mu = \mu$  used everywhere else in Section 4.3. Under that weaker hypothesis, transitivity fails.

**Counterexample.**  $\Sigma = \{a, b, c, d\}$ , Goguen product,  $\mu = 0.5$ ,  $\mathcal{R}(a, b) = \mathcal{R}(c, d) = 0.5$ , all other distinct pairs 0.25. This is a valid T-similarity, and  $\mathcal{R}_{0.5} = \{a, b\}^2 \cup \{c, d\}^2$  is an equivalence. With  $|w| = 2$  the threshold is  $\mu^2 = 0.25$ :  $\mathcal{R}^w(ca, aa) = 0.25 \geq 0.25 \checkmark$ ;  $\mathcal{R}^w(aa, ab) = 0.5 \geq 0.25 \checkmark$ ; but  $\mathcal{R}^w(ca, ab) = 0.125 < 0.25$  (fails). Transitivity fails. (Reflexivity and symmetry do hold.)

**Note.**  $(\mu)^{|w_1|}$  is also ambiguous: ordinary exponentiation, or the  $\otimes$ -power  $\otimes^{|w_1|} \mu$ ? They coincide for the product norm and differ for every other, so this is two readings and two theorems — both false in the instance above.

**Fix.** Strengthen the hypothesis to  $\mu \otimes \mu = \mu$ , at which point  $\mu^{|w|} = \mu$  and the proposition collapses into Proposition 4.1(3) and should be deleted. Or drop it.

#### 25. Conjecture 3.1 and Lemma 3.3 are the same fact with contradictory status [FALSE]

Where: line 266 (Conjecture, in red) and line 315 (Lemma)

**Problem.** Lemma 3.3 is an immediate corollary of the conjecture, and both follow from the same trivial induction. Marking one an open conjecture and the other a settled lemma is inconsistent — and Lemma 3.3 is a proof obligation that must be discharged *before*  $\mathcal{D}_a^{\mu}$  is definable at all, since it is what makes Definition 4.5’s index set finite. As long as the fact it rests on is labelled a conjecture, Definition 4.5 has no right to exist.

**Note.** The conjecture is true and easy. Every clause of Definition 3.5 other than “0 otherwise” forces  $r$  and  $s$  to have the same top-level constructor, and  $\emptyset$ ,  $\varepsilon$  and letters are all separated from each other, so a non-zero value forces a common skeleton by induction. Then  $\{s \mid \mathcal{R}^E(s, r) \neq 0\} \subseteq \{r[b_1, \dots, b_n] \mid b_i \in \Sigma\}$ , of cardinality at most  $|\Sigma|^n$ .

**Fix.** Demote it to a lemma and derive Lemma 3.3 from it. Also fix the typo ( $r_1 = r[a_1, \dots, b_n]$  should end in  $a_n$ ) and the redundant re-quantification (“Let  $r, s$  be restricted ... then for all restricted  $r_1, r_2$ ”).

**Note.** For Rocq, formalize this as a *computable enumeration skeleton\_instances* : `regex -> seq regex`, not as an abstract finiteness statement. That single choice is what turns  $\mathcal{D}_a^{\mu}$  into a genuine function (Phase 5).

#### 26. Theorem 3.1’s “Moreover” clause is malformed; the corrected version is true [FALSE] [UNPROVED]

Where: line 297

**Problem.** “Moreover if  $\mathcal{R}$  is a gödelian T-similarity relation, or  $\mu \otimes \mu = \mu$ , then  $\mathcal{R}^E(r_1, r_2) \geq \mu$  then  $\mathcal{L}^{\mu}(r_1) = \mathcal{L}^{\mu}(r_2)$ ” — two *thens*, no *and*.

**Fix.** “If  $\mathcal{R}$  is gödelian or  $\mu \otimes \mu = \mu$ , and  $\mathcal{R}^E(r_1, r_2) \geq \mu$ , then  $\mathcal{L}^{\mu}(r_1) = \mathcal{L}^{\mu}(r_2)$ .” I checked the corrected statement and it is true under either hypothesis. Two facts it needs, worth recording because Rocq will need them as separate lemmas:

- the supremum in Definition 3.3 is *attained*:  $\Sigma$  is finite and  $\mathcal{R}^w$  vanishes off equal lengths, so the supremum is over a finite set;
- in the gödelian case,  $\otimes$  restricted to idempotents is  $\min$  — if  $a \leq b$  are both  $\otimes$ -idempotent then  $a = a \otimes a \leq a \otimes b \leq \min(a, b) = a$  — so  $\mathcal{R}^w$ - and  $\mathcal{R}^E$ -values inherit idempotence from  $\mathcal{R}$ -values.

**Note.** Give the compound hypothesis a name in Section 3 (for instance, “ $\mathcal{R}$  is  $\mu$ -transitive”). It appears verbatim four times — lines 297, 538, 583, 611 — each time with the dangling “then”.

**27. Theorem 3.1(2)’s hard direction is not known, not flagged, and Admitted in Rocq — with a different statement [UNPROVED] [ROCQ CODE]**

Where: line 292; `coq/src/SimilarityRelations.v:1621 (Rr_RL_bridge, Admitted)`

**Problem.** The  $\leq$  half ( $\sup \leq \mathcal{R}^L$ ) is immediate from part 1. The  $\geq$  half asserts that syntactic similarity can always be pushed up to the semantic one by choosing better representatives — a genuine completeness statement requiring a witness construction, and the only real content of the theorem. It is the one `Admitted` in the whole development, and the text does not flag it as the hard half, so a reader has no way to tell that this is where the risk sits. I could not find a counterexample.

**Note.** The Rocq statement is not the paper’s. What the code proves is `is_lub` for the set  $\{\min(\mathcal{R}^E(r', t), \mathcal{R}^E(s', u)) \mid r' \approx r, s' \approx s, t \approx u\}$ , a strictly *larger* set than the paper’s  $\{\mathcal{R}^E(r'_1, r'_2) \mid \mathcal{L}(r'_i) = \mathcal{L}(r_i)\}$  — take  $t = u = s'$  to recover the paper’s set. Only the upper-bound half is proved (`classical_Rr_candidate_values_upper`); the lower bound is the `Admitted`.

**Fix.** Flag the direction as open in the paper, decide which statement you want, and make the Rocq statement match it. Also say explicitly that the theorem ranges over *restricted* expressions only — the claim would differ over  $E^\Sigma$ .

**28. Remark 4.1’s two derivative counts are both wrong [FALSE]**

Where: lines 432–436

**Problem.** For  $r = a^* + b(a + \varepsilon)^*$  the remark says the semantic set has cardinality two and the syntactic set three. Working it out:

- $\mathcal{D}_\varepsilon(r) = r$ ,  $\mathcal{D}_a(r) \sim a^*$ ,  $\mathcal{D}_b(r) \sim (a + \varepsilon)^*$ ,  $\mathcal{D}_{ab}(r) \sim \emptyset$  — so  $\{\mathcal{D}_u(r) \mid u \in \Sigma^*\}$  has 4 elements up to  $\sim$ ;
- $\partial_\varepsilon(L) = L$ ,  $\partial_a(L) = \partial_b(L) = \{a\}^*$ ,  $\partial_{ab}(L) = \emptyset$  — so  $\{\partial_u(L) \mid u \in \Sigma^*\}$  has 3.

The stated 2 and 3 are the counts over  $u \in \Sigma^+$ , but the sets as written include  $u = \varepsilon$ .

**Fix.** Say 3 and 4, or range over  $\Sigma^+$ . The point of the example — that the syntactic count strictly exceeds the semantic one because  $a^*$  and  $(a + \varepsilon)^*$  are language-equivalent but not similar — is correct and survives either fix. Also, “we have that  $\mathcal{D}_a = a^*$  and  $\mathcal{D}_b = (a + \varepsilon)^*$ ” omits the argument  $r$ , and the cross-reference “Theorem 4.1.3 and .4” should be “.4 and .5”.

**29. Systematic type error: “ $\varepsilon \in \mathcal{D}_u(r)$ ” for “ $\varepsilon \in \mathcal{L}(\mathcal{D}_u(r))$ ”, and  $\mathcal{D}_u(L)$  for  $\mathcal{D}_u(r)$  [FALSE]**

Where: lines 52, 420, 424, 434, 509, 512, 526

**Problem.** A regular expression is not a set. The worst instance is line 52, which states the paper’s central mechanism and gets it wrong: “checking that a word belongs to a regular language is reduced to computing that an iterated derivative *is the empty word*”. The correct statement is that the iterated derivative is *nullable*:  $w \in \mathcal{L}(r) \iff \varepsilon \in \mathcal{L}(\mathcal{D}_w(r))$ .

**Counterexample.** From the paper’s own definitions:  $r = a^*$ ,  $w = a$ . Then  $\mathcal{D}_a(a^*) = \mathcal{D}_a(a) \cdot a^* = \varepsilon \cdot a^*$ , which is nullable but is not the empty word. The sentence also contradicts Definition 4.1, which introduces  $\nu$  for exactly this purpose.

**Fix.** One pass over all seven sites. Theorem 4.2(1) at line 500 already states the fuzzy analogue correctly,  $\varepsilon \in \mathcal{L}(\mathcal{D}_u^\mu(r))$  — copy that form.

**30. Theorem 4.1 carries four labels on one statement, and every reference into it is off by one [FALSE]**

Where: lines 417–418; references at lines 434, 891, 1000, 1088, 1117

**Problem.** Every `\ref` renders “4.1”, so only the hand-written item number distinguishes them — and the item numbers are consistently wrong: `the:language-derivative-regular(1)` is item (2) (regularity); `the:two-derivative-mapping(2)` is item (3) (correspondence); `the:derivative-set-is-finite(3)` is item (5) (semantic finiteness); and “Theorem 4.1.3 and .4” at line 434 should be “.4 and .5”. The same fragility hit Theorem 4.2, whose items silently renumbered when item 4 was commented out.

**Fix.** Split into separately numbered results, or put a `\label` on each item so references cannot drift.

**31. Section 6: the similarity relation  $\mathcal{R}^\lambda$  is not reflexive, so it is not a T-similarity [FALSE]**

Where: lines 712–716 — Section 6 was never reviewed before

**Problem.** The paper defines  $\mathcal{R}^\lambda(a_1, a_2) = \lambda_1 \otimes \lambda_2$  if  $a_1 = a^{\lambda_1}$  and  $a_2 = a^{\lambda_2}$ , and 0 otherwise, and calls it a similarity relation. But  $\mathcal{R}^\lambda(a^{0.5}, a^{0.5}) = 0.25 \neq 1$ , so reflexivity fails. Symmetry and the reverse triangle inequality do hold, but reflexivity is assumed by everything in Sections 3–4.

**Fix.** Use the bi-residuum:  $\mathcal{R}^\lambda(a^{\lambda_1}, a^{\lambda_2}) = (\lambda_1 \Rightarrow \lambda_2) \otimes (\lambda_2 \Rightarrow \lambda_1)$ . That is 1 on the diagonal, symmetric, and satisfies the reverse triangle inequality — and the residuum is already in Appendix A, Definition A.3.

**32. Section 6: the extended alphabet is uncountable, which contradicts Definition 2.1 and every finiteness theorem [FALSE]**

Where: line 711

**Problem.** “we may introduce for each  $a \in \Sigma$  new alphabet symbols  $a^\lambda$  for  $\lambda \in [0, 1]$ ” gives  $\Sigma^\lambda$  the cardinality of the continuum. Definition 2.1 says an alphabet is finite, and Lemma 3.3, Definitions 4.5 and 4.7, and Theorem 4.2(4),(5) all need it.

**Fix.** Restrict  $\lambda$  to a finite set of grades — for instance the finitely many scalars actually occurring in the given fuzzy regular expression  $\alpha$  — and say so.

**33. Section 6: the translation  $r^{\alpha, \lambda}$  is not a definition [BLOCKS ROCQ]**

Where: lines 718–722

**Problem.** The function being defined is  $r^{\alpha, \lambda}$ , but the right-hand sides write  $a_1^{\alpha_1, \lambda}$  —  $a_1$  is never introduced; presumably  $r^{\alpha_1, \lambda}$  was meant. In the second clause, “if  $\alpha = \lambda \alpha_1$ ” produces  $a_1^{\alpha_1, \lambda \otimes \lambda'}$  with  $\lambda'$  unbound. Clauses 1 and 2 overlap, since  $\lambda' a$  is an instance of  $\lambda' \alpha_1$ , so the recursion is not deterministic. There are no clauses for the constants of the scalar regular-expression algebra. And  $\mathcal{RL}$ , the set  $\alpha$  ranges over, is never defined — a commented-out line calls it  $\mathcal{LR}$ .

**Fix.** Rewrite as a total structural recursion on  $\alpha$  covering every constructor, name the scalar-expression syntax, and cite where it comes from in [SC12].

**34. Proposition 6.1 has a free variable and an unused one [FALSE]**

Where: line 723

**Problem.** “for all  $\mu$ , we have that if  $\|\alpha\|(a_1 \dots a_n) \geq \mu$  then there exist  $\lambda_1, \dots, \lambda_n$  such that  $\lambda_1 \otimes \dots \otimes \lambda_n \geq \lambda$ ” —  $\lambda$  is free and should be  $\mu$ ;  $\mu$  is otherwise unused. Also  $\|\cdot\|$  is never introduced.

**Fix.** Replace  $\lambda$  by  $\mu$ , introduce  $\|\cdot\|$ , and state it for restricted  $\alpha$  if that is what is meant.

**35. Proposition 4.5 (red) is true, but its hypothesis names things it never uses [EDITORIAL]**

Where: lines 690–697

**Problem.** It says “let  $\mathcal{R}$  be gödelian . . . moreover let  $\mathcal{L} \subseteq \Sigma^*$  and  $a \in \Sigma$ , then  $\mathcal{L}(\mathcal{D}_a^\mu(r)) \subseteq \{w \mid q_\mu^*(w) \in \mathcal{L}(\mathcal{D}_{[a]_\mu}[r]_\mu)\}$ ” —  $\mathcal{L}$  is never used, and  $r$  is never introduced.

**Note.** I checked the inclusion and it holds. Remark 4.2 does give strictness: with  $\Sigma = \{a, b\}$ ,  $\min$ ,  $\mathcal{R}(a, b) = \mu$ ,  $r = a^*$ , the left side is  $\{a\}^* \cup \{b\}^*$  and the right side is all of  $\Sigma^*$ .

**Fix.** Bind  $r$ , drop  $\mathcal{L}$ , and add the lemma the proof silently uses:  $q_\mu^*(\mathcal{L}(r)) = \mathcal{L}(q_\mu^*(r))$  for restricted  $r$  ( $q_\mu^*$  is a monoid homomorphism and commutes with union, concatenation and star).

**36. Missing lemma: under  $\mu \otimes \mu = \mu$ , Definition 4.6’s union collapses to plain iteration [UNPROVED]**

Where: Definitions 4.6 and 4.7; Section 4.3

**Problem.** This is the bridge to the WiL formulation and to all six dead proofs, and it is stated nowhere.

**Fix.** It is easy:  $\partial_a^{\mu'}$  is antitone in the cut, and  $\otimes \leq \min$  forces  $\mu', \mu'' \geq \mu$ , so the union over  $\mu' \otimes \mu'' \geq \mu$  is attained at  $\mu' = \mu'' = \mu$  when  $\mu \otimes \mu = \mu$ . The same holds for Definition 4.7 up to  $\sim$ , the extra summands being absorbed by idempotence of  $+$ . State it as a lemma in Section 4.3: it is what lets Section 4.3 reuse the WiL proofs, and what lets the fuzzy DFA reduce to the crisp one over  $\Sigma^\mu$ .

**37. Missing lemma: gödelian  $\Rightarrow \mathcal{R}^w, \mathcal{R}^E, \mathcal{R}^L$  take only  $\otimes$ -idempotent values [UNPROVED]**

Where: Proposition 4.1, line 538

**Problem.** Proposition 4.1(2),(3),(4) claim that the  $\mu$ -cuts of  $\mathcal{R}^E, \mathcal{R}^w$  and  $\mathcal{R}^L$  are equivalences under “ $\mathcal{R}$  gödelian”. But gödelian is a hypothesis about the values of  $\mathcal{R}$  on  $\Sigma \times \Sigma$ , so it says nothing directly about the derived relations. Without the transfer lemma the gödelian half of Proposition 4.1 does not follow.

**Fix.** Add it; it is short. On idempotents  $\otimes = \min$  (item 26);  $\min$ ,  $\inf$  and  $\sup$  of idempotents are idempotent; and  $\inf_n \otimes^n x = x$  for idempotent  $x$ . Since  $\Sigma$  is finite,  $\mathcal{R}$  takes finitely many values, so  $\mathcal{R}^w, \mathcal{R}^E$  and  $\mathcal{R}^L$  take values in the finite set of  $\mathcal{R}$ -values together with 0 and 1 — all idempotent.

## Part D — Smaller errors and editorial, in one pass

### 38. Wrong reference for “the standard theory of finite automata” [EDITORIAL]

Where: line 36, \cite{C71}

C71 is Conway, *Regular Algebra and Finite Machines* (1971) — a monograph on regular algebra, not a standard finite-automata reference. Cite a textbook, or drop the citation.

### 39. Wrong reference for the fuzzification of automata theory [EDITORIAL]

Where: line 48, \cite{C71}

Conway’s book contains no fuzzy automata at all. Cite LP06 (already in the bibliography), or Mordeson–Malik, Wee67, WeeFu69. Note that C71 *is* appropriate elsewhere — it is where Redko’s theorem is reported — so in effect C71 and the fuzzy-automata references are swapped relative to where they belong.

### 40. Category error: membership functions versus lattice-ordered monoids [EDITORIAL]

Where: lines 48 and 706

“by turning membership functions into lattice-ordered monoids”. A membership function *takes values in* one. Same slip at line 706 (“fuzzy sets as functions in some lattice-ordered monoid”). Fix: “by letting membership functions take values in a lattice-ordered monoid (or a residuated lattice) instead of in  $\{0, 1\}$ ”.

### 41. B62 versus B64 for “Brzowski’s system” [EDITORIAL]

Where: line 101; biblio.bib:94

Line 101 attributes the equational system to B62, but the ACI-based similarity and the finiteness-of-derivatives result are in the 1964 JACM paper, which the abstract and introduction already cite. Also the B62 entry is typed @article with journal “Mathematical Theory of Automata”; it is a symposium proceedings (MRI Symposia Series vol. 12, Polytechnic Press). Fix both.

**Note.** The related citation problem from the earlier notes is *already fixed*: line 101 now cites Redko for the negative result and correctly describes Salomaa and Kozen as not purely equational.

### 42. Left-continuity is assumed throughout and never defined [BLOCKS ROCQ]

Where: line 744

It is load-bearing in three places: Proposition A.1(1) (that  $\otimes$  *uniquely determines* a residuum); Lemma 3.2(3) (the reverse triangle inequality for  $\mathcal{R}^L$  needs  $\otimes$  to commute with sup — this is the one place it genuinely bites in Section 3); and Definition 3.5’s star clause. Fix: add to Definition A.2 — “ $\otimes$  is *left-continuous* if  $\sup_i (x \otimes y_i) = x \otimes \sup_i y_i$  for every  $x$  and every increasing chain  $(y_i)$ ” — and say in Proposition A.1 that item (1) is exactly where it is used.

### 43. $\mu_S$ (membership function) collides with $\mu$ (the cut value) [EDITORIAL]

Where: lines 732–736

\cut is \def’d to  $\mu$  and denotes the cut value everywhere else; the appendix uses  $\mu_S$  for the membership function. Two different  $\mu$ ’s, one of them the paper’s central parameter. Rename the membership function ( $m_S$  or  $\chi_S$ ).

#### 44. Appendix A duplicates Section 3’s fuzzy-relation definitions [EDITORIAL]

Where: Definition A.5 (line 809) vs. the unnumbered Definition 3.1 (line 185)

Both define fuzzy relation,  $\mu$ -cut, T-similarity and extensional. Keep one. Notes for deciding: the appendix copy has zero cross-references, omits *crisp*, *gödelian* and *locally idempotent at  $\mu$*  — all three introduced only inline in Section 3, even though line 182 sends the reader to the appendix for “standard notions” — while the body copy is used in the very next paragraph, so keeping only the appendix copy would make the body point forward for a notion it uses immediately. Also `\label{def:fuzzy-relation}` is never referenced.

#### 45. The 1-cut is written $\mathcal{R}^1$ where everything else writes $\mathcal{R}_\mu$ [EDITORIAL]

Where: line 203; line 536 writes  $\mathcal{R}^\mu$  and  $\mathcal{R}_\mu$  in the *same* sentence

Unify to subscripts.

**Note.** The claim itself is correct: the 1-cut is an equivalence for every T-norm (since  $1 \otimes 1 = 1$ ), and  $\mathcal{R}$  is well defined on its classes — if  $\mathcal{R}(s, s') = 1$  then  $\mathcal{R}(s, t) \geq 1 \otimes \mathcal{R}(s', t) = \mathcal{R}(s', t)$  and symmetrically.

#### 46. The extensionality assumption is stated twice in the same paragraph [EDITORIAL]

Where: line 203, first and last sentence

“Throughout the paper we assume that T-similarity relations are extensional . . . Hence throughout the paper we will assume that T-similarity relations are always extensional.” Drop one.

#### 47. “Gödelian” is a property of a relation, but line 204 applies it to a norm [EDITORIAL]

Where: line 204

“Gödel’s minimum T-norm is obviously gödelian.” Under min every value is idempotent, so every T-similarity is gödelian with respect to min — say that instead.

#### 48. The ordinal sum is named in one order and constructed in the other [EDITORIAL]

Where: line 205 vs. Definition A.4(4), line 771

Line 205 says “the  $p$ -ordinal sums of Gödel’s and Goguen’s T-norms”; Definition A.4(4) says “the  $p$ -ordinal sum of Goguen’s product and Gödel’s minimum”. The construction puts min below  $p$  and the product above, so the increasing-interval order is “Gödel’s minimum below  $p$ , Goguen’s product above”. Use it consistently.

**Note.** I checked Definition A.4(4)’s formula *and* its residuum, and both are correct; so is the claim at line 205 that these norms are idempotent on  $(0, p]$ .

#### 49. Definition 3.4 writes $T$ where the paper writes $\otimes$ [EDITORIAL]

Where: line 237

“Let T be Gödel’s T-norm”.

#### 50. Definition 3.5 has no signature, and its first clause overlaps the others [EDITORIAL]

Where: line 255

Unlike Definition 3.4 it never declares its domain and codomain. And “1 if  $r = s$ ” overlaps every following clause — the values agree (I checked), so it is harmless, but the definition should say the cases are tried in order, or drop the first clause, which is derivable.

## 51. $E^\Sigma$ versus $E$ [EDITORIAL]

Where: Definition 2.1 (line 73) vs. lines 238, 257

Definition 2.1 introduces  $E^\Sigma$  for regular expressions and  $R^\Sigma$  for restricted ones; Section 3 writes  $\mathcal{G}^E : E \times E \rightarrow [0, 1]$  and  $\mathcal{R}^E$  over an unqualified  $E$ . Pick one.

## 52. Three unfinished sentences [EDITORIAL]

Where: lines 58, 441, 536

(a) Line 58: the Synopsis reads `In Section~\ref{basic} \ldots` and stops there. (b) Line 441: “Our goal are fuzzy versions of so as to achieve the fuzzy version of Theorem 4.1, namely Theorem 4.2 and” — a hole after “versions of” and a break after “and”. (c) Line 536: “Capitalizing on this all definitions and theorems of Section 4.2 because all  $\mu$ -neighborhoods ... become equivalence classes” — no main verb.

## 53. Definition 4.8’s quotient language is nested inside the wrong item [EDITORIAL]

Where: lines 624–631

The definition of  $[L]_\mu$  sits inside item 2, which is about regular expressions. Make it its own item.

## 54. “If $\Sigma$ is finite” is redundant everywhere it appears [EDITORIAL]

Where: Theorem 4.1 (line 419), Theorem 4.2(1),(4),(5), Lemma 3.3 (line 315)

Definition 2.1 already says an alphabet is a *finite* nonempty set. Either drop the hypotheses, or drop finiteness from Definition 2.1 and keep them.

**Note.** Do not conclude the hypothesis is *vacuous*: Definition 4.5’s sum over  $b \in \Sigma$  genuinely needs it. It is redundant, not unnecessary.

## 55. Appendix B’s substitution example is off by one, and the appendix is orphaned [EDITORIAL]

Where: lines 833–844

With  $r[x] = a \cdot x \cdot b$  (written “ $r = a[a]b$ ”) and the stated recursion,  $r_0^\# = a \cdot a \cdot b \mapsto \{aab\}$  and  $r_n^\# \mapsto \{a^{n+2}b^{n+1}\}$ , so  $\mathcal{L}(r[a]^\#) = \{\varepsilon\} \cup \{a^{m+1}b^m \mid m \geq 1\}$ . The paper writes  $\{\varepsilon\} \cup \{a^{n+1}b^n \mid n \in \mathbb{N}\}$ , which — if  $\mathbb{N}$  contains 0 — wrongly includes the single letter  $a$ . The conclusion (not regular) is correct either way. Also: nothing in the main text references Appendix B; `\label{extension}` is attached to a bold run of text rather than a sectioning command, so it resolves to “B”; the notation  $a[a]b$  is not the  $r[a_1, \dots, a_n]$  notation of Section 2; and  $r[a]^\#$  is only ever given a language, never defined as an expression. Fix: shift the exponents, wire the appendix into Section 2 or drop it, and reconcile the notation.

## 56. Section 5, “Formalization in Rocq”, is an empty section heading [EDITORIAL]

Where: lines 701–703

It is followed immediately by Section 6. Since the paper’s selling point is that the results are machine-checked, this is the section a referee will look for first.

## 57. “(tentative)” appears in a section title [EDITORIAL]

Where: line 704

**58. Placeholder banners and 18 red passages survive into the compiled PDF [EDITORIAL]**

Where: lines 36 (“TO BE REWRITTEN”), 47 (“TO BE COMPLETED”), and red markers at 214, 266, 276, 296, 319, 432, 520, 526, 565, 568, 608, 658, 690

Six of them are “Rocq proof ??” / “Rocq?? Proofs??” placeholders, standing where a Rocq citation will go, on Lemma 3.1, Lemma 3.2, Theorem 3.1, Lemma 3.3, Proposition 4.1 and Proposition 4.3 — a useful working checklist of exactly which results still need one. Keep them in the working copy; hide them, and the two banners, before the document is circulated.

**59. “Most of our results are formalized in Rocq” is not true [FALSE]**

Where: lines 42 and 54

README.md’s mapping table has 20 todo rows; the entire fuzzy half is unformalized; the one formalized fuzzy theorem is `Admitted`; and two source files do not compile (item 64). Fix: “The classical results are formalized in Rocq; the fuzzy development is formalized in part.”

**Note.** The companion claim “following the approach of Coquand and Siles” does not describe the code either — `coq/src/` is a boolean/ssreflect development, not their decision procedure.

**60. Four bibliography entries are never cited, including the one that matters most [EDITORIAL]**

Where: `tex/sections/biblio.bib`

`Glushkov1962`, `McNaughtonYamada1960`, `Thompson1968` and `RocqRegexpBrzozowski` are defined but uncited. The last is an existing Rocq formalization of Brzozowski derivatives and belongs in Section 6: the paper should say how this development relates to it.

**61. KM26 is cited both as a “special case” and as something “generalized” [EDITORIAL]**

Where: lines 46 and 534

Line 46 calls the WiL abstract “the special case of the present work for the minimum T-norm”; line 534 says Section 4.3 “is a generalization of the results in [KM26]”. Both cannot stand once item 2 is settled.

**62. The nf-based repair suggested at line 312 is contradictory, not merely unattractive [FALSE]**

Where: line 312

The text offers “we could put  $\mathcal{R}^E(r_1, s) = \mathcal{R}^E(r_2, s)$  if  $\text{nf}(r_1) = \text{nf}(r_2)$ ” as a workable-but-incomplete option. It is inconsistent. Since  $\text{nf}(a + b) = \text{nf}(b + a)$ , the constraint forces  $\mathcal{R}^E(a + b, a + b) = \mathcal{R}^E(b + a, a + b)$ , that is  $1 = \mathcal{R}(a, b)$ , for *every* pair of symbols. To be a definition it must say which value to take on each  $\sim$ -class; the natural choice is the supremum over the class, which is precisely Theorem 3.1(2) with semantic equivalence replaced by  $\sim$ .

**63. Remark 2.1’s claim that nf decides  $\sim$  leans on the equational reading [EDITORIAL]**

Where: line 101

The remark’s point — “a finite approximation of equality is enough to achieve finiteness in derivatives and hence in states” — is fine, but it presupposes Lemma 2.1 in its equational reading. Re-check after item 4. And note that item 5 shows the states part needs care.

## Part E — Problems in the existing coq/ development

### 64. The build is broken: two files import a module that was deleted [ROCQ CODE] [BLOCKS ROCQ]

Where: `coq/src/AutomataConstruction.v:4` and `coq/src/AutomataExample.v:5`

**Problem.** Both `do From regexderiv Require Import FiniteAlphabet.` — but that file does not exist. Commit `c083627` (“remove `FiniteAlphabet.v`”) deleted it without removing the imports, and `_CoqProject` does not list it. Since `_CoqProject` maps `-Q src regexderiv`, the import can only resolve to `src/FiniteAlphabet.v`, so `make` cannot get past `AutomataConstruction.v`. Its contents were folded into `Alphabet.v` as the module types `FINSYM` and `OFINSYM`.

**Fix.** Delete both import lines, then check that `alphabet`, `alphabet_uniq`, `alphabet_nonempty` and `alphabet_complete` all resolve through `OFINSYM`. Nothing else should be needed. Do this first: until it is done, no claim about what the development proves can be checked.

### 65. The functor tower creates several incompatible copies of the same regex type [ROCQ CODE]

Where: `Nullable.v:13`, `Canonicalization.v:13`, `SimilarityRelations.v:17`; consequences at `Automata.v:21–67` and `AutomataConstruction.v:21–44`

**Problem.** `Nullable X`, `Canonicalization X` and `SimilarityRelations X` each apply `RegexSemantics X` separately, so `Rocq` generates *distinct* inductive types all called `regex`. That is why `Automata.v` and `AutomataConstruction.v` each carry a 45-line `to_c_regex/from_c_regex` translation plus two correctness lemmas — pure boilerplate to move between two copies of the same definition. The fuzzy layer is worse: the `regex` in `SimilarityRelations.v` is a third copy, so `Rr` cannot be applied to a `D_char` output without yet another translation. This is the structural reason the fuzzy half has never been connected to the derivative half.

**Fix.** Drop the functors. In `mathcomp` the idiomatic form is a section variable: `Section Dev. Variable A : finType.` — one `regex` type, no translations, and `finType` hands you the finite enumeration, decidable equality and finite-quantification lemmas that every finiteness theorem in the paper needs. The order that canonicalization requires is derivable (`cmpA a b := index a (enum A) ?= index b (enum A)`), so `Alphabet.v`’s four module types collapse to nothing. If the functors must stay, at least have downstream modules take the *instantiated* upstream module as a parameter (`Module Automata (X : OSYM) (C : CANON X)`) instead of re-applying it.

**Note.** This is the single change that will most reduce the cost of the fuzzy half. Do it before Phase 3, not after.

### 66. The fuzzy layer is written for the minimum T-norm [NOT YET DONE]

Where: `SimilarityRelations.v:26, 120, 143, 149, 1125, 725`

This is the code side of item 1, and it is outstanding work rather than a defect. `smin` occurs in `is_similarity`, in the hypothesis `R_trans`, and in the definitions of `Rw`, `Rr` and `RL` — that is, in the *statements*, which is what makes the port a restatement. Fix: the T-norm interface of Phase 3, with `smin` as its first instance, then Phase 4.

**67. Rr is defined on And and Neg — the extension the paper refutes** [ROCQ CODE] [FALSE]

Where: SimilarityRelations.v:1155–1166

$Rr (And\ r1\ r2) (And\ s1\ s2) = smin\ (Rr\ r1\ s1)\ (Rr\ r2\ s2)$  and  $Rr\ (Neg\ r1)\ (Neg\ s1) = Rr\ r1\ s1$ . Definition 3.5 has no such clauses, and Section 3’s counterexamples exist precisely to show those clauses break Theorem 3.1(1) (item 22). Fix: restrict Rr to the restricted fragment, or keep the total version under a different name and state clearly that it is not Definition 3.5.

**68. Two identical copies of the restricted fragment, and neither is used where it matters** [ROCQ CODE]

Where: RegexSemantics.v:25 (restricted\_regex) and SimilarityRelations.v:1287 (classical\_regex)

**Problem.** restricted\_regex is defined and then *never mentioned again* anywhere in the development. classical\_regex is the same six constructors under a different name, and it is the one used as the hypothesis of the bridge lemmas. So the paper’s central side condition exists twice and is enforced in one place only.

**Fix.** One notion, used everywhere the paper says “restricted”. Two workable shapes: (a) a boolean predicate is\_restricted : regex -> bool plus closure lemmas (is\_restricted r -> is\_restricted (D\_char a r), which is easy once nu r is known to be Eps or Empty) — cheap, and it fits the existing code; (b) a separate inductive type rregex with an embedding into regex — every fuzzy signature then becomes honest by construction and no closure lemma is needed, at the cost of duplicating the derivative. Recommended: (a).

**69. The only formalized fuzzy theorem is Admitted, and it is not the paper’s theorem** [ROCQ CODE] [UNPROVED]

Where: SimilarityRelations.v:1621

See item 27. README.md nonetheless lists it as formalized (item 76).

**70. The equational theory of Definition 2.2 is not formalized at all** [ROCQ CODE]

Where: Canonicalization.v:215

See item 4. There is no inductively defined similar anywhere in coq/src/, so similar\_iff\_nf\_eq is a tautology and the paper’s Lemma 2.1 has no machine-checked content. What *is* genuinely proved is canonize\_correct (canonize preserves the denotation) and hence similar\_sound — that is Lemma 2.2, and it is solid.

**71. Automata.v’s DFA has all regexes as states, so nothing about finiteness is proved** [ROCQ CODE] [UNPROVED]

Where: Automata.v:116 (constructed\_dfa), 149 (dfa\_correctness)

**Problem.** constructed\_dfa sets  $Q := \text{regex}$ . So dfa\_correctness says “this deterministic transition system recognises  $\mathcal{L}(r)$ ” — true, cleanly proved, and useful — but it is not the paper’s theorem, whose entire content is that  $Q = \{\mathcal{D}_u(r)\}$  is *finite*. Brzowski’s finiteness theorem (Theorem 4.1(4)) appears nowhere in coq/src/.

**Note.** Because  $Q$  is all of regex, the two well-definedness obligations the paper’s construction needs — “ $\mathcal{D}$  respects  $\sim$ ” and “ $\nu$  respects  $\sim$ ” — never arise. The first of them is **false** (item 5), so they cannot simply be bolted on later.

**Fix.** See Phase 2.

## 72. The executable construction is unverified and its termination is assumed [ROCQ CODE] [UNPROVED]

Where: AutomataConstruction.v:200 (construct\_dfa), 218 ((\*TODO L(M) = L(r)\*)

**Problem.** `construct_dfa` (`fuel` : `nat`) (`r` : `regex`) explores the reachable state set with a structurally decreasing `fuel` counter, and its correctness is a comment. `fuel` stands in for the termination argument that would follow from finiteness of the reachable set — which is precisely what is not proved, and which item 5 shows is not the same set as  $\{\mathcal{D}_u(r)\}$ .

**Fix.** Make it return `option dfa` (`None` when `fuel` runs out) and prove the honest statement: if it returns `Some M` then `dfa_lang M`  $\equiv$  `den r` and `dfa_well_formed M` holds. That is fully verifiable today, it is a real result, and it removes the pretence that termination has been established. Prove termination separately, later, if at all.

## 73. SupL and InfL carry proof arguments, and the friction propagates through the whole fuzzy layer [ROCQ CODE]

Where: SimilarityRelations.v:281 and 349; consequences at 234–700

**Problem.** `SupL` (`L` : `language`) (`f` : `word -> simval`) (`Hub` : `upper_bounded_on L f`) makes every value depend on a *proof term*. That is why the file needs `SupL_eq_ext`, `InfL_eq_ext`, `SupL_upper`, `SupL_least_nonempty`, `upper_bounded_on_smin_const` and about twenty more, and why every statement has to thread matching proofs (see the `assert (Heq : ...)` block in `syntax_semantics_inequality`). But all values lie in  $[0, 1]$ , so *every* such family is bounded — the hypothesis is never informative.

**Fix.** Define total operators `Sup01` (`P` : `R -> Prop`) : `R` and `Inf01`, with the empty conventions (0 and 1) built in, and prove the `lub/glb` characterisations as lemmas. This removes roughly the entire 450-line plumbing section and makes every downstream statement readable.

## 74. The fuzzy layer is classical and non-computational; the crisp layer is boolean and executable [ROCQ CODE] [DECISION]

Where: SimilarityRelations.v:2 (`Reals`, `Lra`, `ClassicalEpsilon`, `FunctionalExtensionality`); `excluded_middle_informative` at 283, 351, 1518

**Problem.** The crisp layer (`Languages.v`, `Derivatives.v`, `Canonicalization.v`) is `bool`-valued and computes; the fuzzy layer is classical. Since  $\mathcal{D}_a^\mu$  must be *computable* for Theorem 4.2(1) to be a decision procedure, this seam is exactly where the fuzzy half will break.

**Fix.** Decide deliberately in Phase 3: keep the reals for the value algebra, but expose a *decidable* comparison  $\mu \leq \mathcal{R}^E(s, r)$  as an interface obligation, so that  $\mathcal{D}_a^\mu$  can be defined by `filter` rather than by `comprehension`. See F2.

## 75. ExamplesCanonicalize.v pins a mathcomp-internal generated name [ROCQ CODE] [EDITORIAL]

Where: ExamplesCanonicalize.v:14

**Definition A** : `eqType` := `Datatypes_bool__canonical__eqtype_Equality`. This name is produced by `mathcomp`'s canonical-instance generator and changes between versions, so the file will break on any `mathcomp` upgrade. **Fix:** `Definition A` : `eqType` := `bool` and let canonical structures do the work.

## 76. README.md’s mapping table is out of date, and several rows are wrong [ROCQ CODE] [FALSE]

Where: README.md

- It calls `tex/` “the mathematical paper”, and every “Paper location” cell points at `tex/sections/*.tex` — the *oldest* version. CLAUDE.md says `dhkm.tex` is canonical.
- It lists `Rr_RL_bridge` as formalized; it is Admitted.
- It lists `same_after_canon` and `canonize_equiv` as Rocq items. `same_after_canon` exists only as an example helper in `ExamplesCanonicalize.v`; `canonize_equiv` does not exist at all.
- It has a row for a “DFA minimality theorem”, which `dhkm.tex` does not contain.
- One row is malformed — the `Rr_RL_bridge` row runs into the next one, with a `|` and a row title missing.
- It maps “Derivative DFA states  $[\mathcal{D}_u(r)]$ ” to `Automata.v/constructed_dfa`, which is not what that code says (item 71).

**Fix.** Regenerate the table against `dhkm.tex`’s numbered results — one row per numbered result, with an explicit status: *proved* / *admitted* / *todo* / *not in the paper*.

## 77. Small tidying in `coq/` [ROCQ CODE] [EDITORIAL]

Where: various

- `Derivatives (X : OSYM)` never uses the ordering; `SYM` would do. Same for `Automata (X : OSYM)` apart from what it inherits from `Canonicalization`.
- `_CoqProject` lists `Languages.v` before `Alphabet.v`; harmless, since `Languages.v` does not depend on it, but the file order no longer reflects the dependency order the way CLAUDE.md claims.
- `Derivatives.v` re-declares `Definition language := pred word`, shadowing the one from `RegexSemantics`.
- `counterexamples.txt` records exactly the two star examples that matter ( $a^*$  vs  $\varepsilon + aa^*$ ,  $a^*$  vs  $(\varepsilon + a)^*$ ) and then trails off mid-sentence. Worth turning into a comment in `Canonicalization.v`, next to `mkStar`, where the decision it records actually lives.

## Part F — What cannot be formalized yet, and why

These are not on the numbered list, because they are not defects to fix one by one — they are places where formalization is blocked by something other than an edit. Referenced above as **F1–F10**.

**F1. Theorem 4.2 as a whole.** Until items 6 and 7 are fixed, Definitions 4.6 and 4.7 do not denote anything, so none of the five statements is a statement. This is the only true hard stop: everything else in Part F is a matter of scope or of research, but this one is a matter of the text not yet saying anything.

**F2. Theorem 4.2(1) as a *decision procedure*.** Definition 3.5’s star clause is  $\inf_n \otimes^n \mathcal{R}^E(r_1, s_1)$ , an infimum over a countable descending chain. The infimum exists —  $[0, 1]$  is complete — so the definition is fine, and this is *not* a defect in the paper. But it forces two choices in the development, and it means  $\mathcal{R}^E(s, r) \geq \mu$  is **not decidable** in general, so Definition 4.5’s index set is a classical set rather than a computable filter. Options:

- add a closed form for the star clause to the T-norm interface — a parameter  $\omega(x) = \inf_n \otimes^n x$  with axioms. For min,  $\omega = \text{id}$ ; for the Goguen product and for Łukasiewicz,  $\omega$  is the indicator of  $\{1\}$ ; for the  $p$ -ordinal sum,  $\omega(x) = x$  for  $x \leq p$  and  $p$  for  $x > p$ . This is the recommended route: it keeps the derivative computable and it is honest about what has been assumed;
- or accept that the *experimentum crucis* is extensional only, and say so in the paper — in which case the phrase “experimentum crucis” should be justified differently.

The truth-value type must also carry chain infima. `Coq.Reals` does (via `completeness`, which `SimilarityRelations.v` already uses), so no change is forced — but see F10.

**F3. Theorem 4.2(2), (4) and (5) for a general norm.** None of the three is known to hold, and the only existing arguments assume  $\mu \otimes \mu = \mu$ . Regularity in particular reduces to a threshold cut of a max- $\otimes$  weighted automaton (item 18), which is a substantial theorem in its own right, and whether the two finiteness results survive without idempotence is open. Formalize all three under  $\mu \otimes \mu = \mu$ ; leave the general case flagged as open in the paper. Do not budget for them.

**F4. Theorem 3.1(2)’s  $\geq$  direction.** Admitted in Rocq, and the Rocq statement is not the paper’s (item 27). The *statement* is formalizable today — it is a supremum over a predicate on expressions, which `Sup01` handles — but nobody has a proof of it. Keep it `Admitted` and label it as such in `README.md`.

**F5. The fuzzy DFA.** Cannot be formalized, because it is not defined. The dead block’s theorem at line 1124 says “the DFA constructed from  $r$  using fuzzy derivatives” and never gives  $Q^\mu$ ,  $\delta^\mu$ ,  $q_0^\mu$  or  $F^\mu$ ; its proof routes through the false equality of item 19; finiteness of  $Q^\mu$  is never asserted; and it introduces a *third* notation,  $\mathcal{L}_\Sigma^\mu(r) := (q_\mu^*)^{-1}(q_\mu^*(\mathcal{L}(r)))$ , for what is — under idempotence — just  $\mathcal{L}^\mu(r)$ . Define it before promising it.

**F6. The paper’s crisp DFA, as defined.** Ill-formed, because  $\mathcal{D}$  is not a congruence for  $\sim$  (item 5). What *is* formalizable is either the ACI-only version (where Brzozowski’s theorem applies verbatim) or the reachable-closure version (where finiteness becomes a new obligation).

**F7. Completeness of nf for Definition 2.2’s equations.** Formalizable, but genuinely new work: an inductive `similar`, plus a termination and local-confluence argument for the rewrite system that `canonize` implements. And item 5 means it does not buy you the automaton, so it is worth doing only if you take reading (i) of item 4.

**F8. Section 6’s comparison with the scalar approach.** Proposition 6.1 and the translation  $r^{\alpha,\lambda}$  are not statements yet (items 31–34), and the extended alphabet is uncountable, which contradicts every finiteness theorem. Fix the text first; the mathematics may well be fine.

**F9. Appendix B’s # operator.** Formalizable but pointless: it deliberately leaves the regular languages, nothing else in the paper uses it, and it is not referenced from the main text.

**F10. The choice of real-number library.** `SimilarityRelations.v` currently mixes `Coq.Reals` (plus `ClassicalEpsilon` and `FunctionalExtensionality`) with `mathcomp ssreflect`. That works, but it is the reason for the `excluded_middle_informative` scattered through the file and for the proof-carrying suprema (item 73). If left-continuity and chain infima are going to be stated properly, `mathcomp-analysis` is a better fit and composes with the rest of the development. This is a dependency decision you cannot easily reverse later — make it before Phase 3, not during it.

## Part G — A plan for formalizing the whole paper

Phases are ordered by dependency. For each, “needs from the paper” lists the numbered items that must be settled first — if they are not, the phase cannot start, and no amount of Rocq skill will help.

### Phase 0 — Repair, and re-lay the foundation

*Small. Do it first, whatever you decide about the paper.*

- Delete the two `FiniteAlphabet` imports; confirm `make completes` (item 64).
- Replace the `SYM/OSYM/FINSYM/OFINSYM` tower with a single `Variable A : finType` in a section-based development; delete `to_c_regex`, `from_c_regex` and their four correctness lemmas (item 65). This is the highest-leverage change in the whole plan.
- One restricted-fragment notion, `is_restricted : regex -> bool`, with the closure lemmas for `D_char` and `nu` (items 68 and 15).
- Rename or restrict `Rr` so that it matches Definition 3.5 (item 67).
- Regenerate `README.md`’s table, one row per numbered result, with statuses (item 76).

**Needs from the paper:** items 8 (domain of  $\mathcal{R}^E$ ), 15 ( $\nu$ ’s clause), 4 (what  $\sim$  is).

### Phase 1 — Crisp core

*Mostly already done and in good shape.*

- Keep `Languages.v`, `Nullable.v`, `Derivatives.v` and `Canonicalization*.v`. `D_char_correct`, `D_correct`, `canonize_correct` and `derivative_regular` are solid and need nothing.
- Add: `nf` is idempotent;  $\nu$  respects  $\sim$ ; and — as an explicit example, so that the design decision is recorded in the code rather than rediscovered — the counterexample of item 5 showing that `D_char` does *not* respect  $\sim$ .
- Introduce `nf_aci` (ACI of  $+$  only) alongside `nf`, and prove that `D_char` is a congruence for `nf_aci`. This is what the automaton will use.

**Needs from the paper:** item 5 (which equivalence the automaton uses).

### Phase 2 — Finiteness and the crisp automaton

*This is the crisp gap, and every finiteness result in the paper — fuzzy included — routes through it. Two routes; pick one now.*

- **Cheap and honest (about a week).** Change `construct_dfa` to return `option dfa` and prove: if it returns `Some M` then `dfa_lang M  $\equiv$  den r` and `dfa_well_formed M`. No finiteness needed, nothing is assumed, and the automata claims in the abstract are backed by something machine-checked (item 72).
- **Full (weeks, not days).** Prove Brzozowski’s finiteness theorem for `nf_aci`: every  $\mathcal{D}_u(r)$  is an ACI-sum of concatenations of subterms of  $r$  and stars of subterms, so  $Q = \{\mathcal{D}_u(r)\}$  is finite and `construct_dfa` terminates. This is the linchpin — Theorem 4.1(4), Theorem 4.2(4),(5) and both DFA theorems all need it.

Recommended: do the cheap route now, and schedule the full route as its own milestone rather than as a step inside the fuzzy work.

**Needs from the paper:** items 5, 30 (fix the references into Theorem 4.1 so the dependency is legible).

### Phase 3 — The value algebra

*The fuzzy foundation, and the first half of the port in item 1.*

- A `Record tnorm` over  $\mathbb{R}$ : commutative, associative, monotone, 1 neutral, left-continuous — the last stated as the sup-preservation equation of item 42.
- Total `Sup01` and `Inf01`, with `sup  $\emptyset$  = 0` and `inf  $\emptyset$  = 1` built in; retire the proof-carrying `SupL/InfL` (items 73 and 12).
- `tpow : nat -> R -> R` (the  $\otimes$ -power) and `omega x := Inf01 {tpow n x}`, with the closed forms for the four named norms (item 12, F2).
- A decidability obligation on the interface — `forall mu x, {mu <= x} + {not (mu <= x)}` over the values actually used, or a concrete `omega`. This is what makes  $\mathcal{D}_a^\mu$  computable (item 74, F2).
- Instantiate: min, Goguen product, Łukasiewicz,  $p$ -ordinal sum.

**Needs from the paper:** items 12 (conventions), 42 (left-continuity), and the F2 decision. **Needs a dependency decision:** F10.

### Phase 4 — Fuzzy relations over a general norm

- `Rw`, `RL` and `Rr` ( $= \mathcal{R}^E$ ) re-stated over the `tnorm` interface. Port the existing min-only proofs `Rw_is_similarity`, `RL_is_similarity` and `Rr_is_similarity`: the proof structures survive, and `smin` becomes `tnorm` in the  $\otimes$  positions while staying min in the  $+$  clause of  $\mathcal{R}^E$  and in the outer combination of  $\mathcal{R}^L$ .
- `syntax_semantics_inequality` = Theorem 3.1(1). Port it: I verified the argument goes through for a general norm, clause by clause.
- The gödelian transfer lemma (item 37) and the “ $\otimes$  is min on idempotents” lemma (item 26).
- `Rr_RL_bridge`: state whichever version the paper settles on, keep it `Admitted`, and say so in `README.md` (item 27, F4).

**Needs from the paper:** items 7, 9, 10, 26, 37.

### Phase 5 — Make the fuzzy syntactic derivative a real function

*This is the pivot of the whole fuzzy half. Everything downstream is easy once it exists, and impossible until it does.*

- `symbol_count r : nat` and `skeleton_instances r : seq regex` — all  $r[b_1, \dots, b_n]$  with  $b_i$  drawn from `enum A`. Then prove `Rr s r <> 0 -> s \in skeleton_instances r`. That is Conjecture 3.1 and Lemma 3.3, made constructive (item 25).
- `D_fuzzy (mu : R) (a : A) (r : regex) : regex` — fold `Alt` over the filter of `[seq (b,s) | b <- enum A, s <- skeleton_instances r]` by the decidable test, in the enumeration order (items 11 and 13).
- `D_fuzzy_word` over the *finite* set of critical cut values (item 6).

- Free consequences worth recording:  $D\_char$  a  $r$  is always a summand;  $D\_fuzzy$  is antitone in  $\mu$ ; the index set is never empty.

**Needs from the paper:** items 6, 11, 13, 25.

## Phase 6 — Fuzzy semantics and the inclusion chain

- $dlang\_fuzzy$  (Definition 4.4) and  $Lmu$  (item 7), with the  $\partial_\varepsilon^\mu(L) = L^\mu$  base case.
- $\mathcal{L}(\mathcal{D}_a^\mu r) \subseteq \partial_a^\mu(\mathcal{L}r) \subseteq \mathcal{L}^\mu(\mathcal{D}_a^\mu r)$  (item 19). Both halves are short; I have checked both proofs.
- The separating counterexample of item 19 as a concrete lemma over a three-letter alphabet — it is the kind of thing that is much more convincing machine-checked than on paper.
- Theorem 4.2(1), the *experimentum crucis* (item 17), once Phase 5 makes  $\mathcal{D}_u^\mu$  computable.

**Needs from the paper:** items 6, 7, 17, 19.

## Phase 7 — The idempotent case: reduce to the crisp development

*This is where the payoff is cheapest, because it **reuses** Phases 1–2 instead of redoing them. If time is short, this is the phase to aim at.*

- Hypothesis `mu_idem` : `tnorm mu mu = mu`, local, never global.
- Proposition 4.1 (all four cuts are equivalences) and Proposition 4.3 (the product condition splits into two separate conditions) — both short, given item 37.
- The collapse lemma (item 36): under `mu_idem`, Definitions 4.6 and 4.7 are plain iteration. This is what lets the WiL proofs be reused.
- The quotient alphabet  $\Sigma^\mu$  as a `finType` — `mathcomp` has quotients of a `finType` by a decidable equivalence — plus  $q_\mu$ ,  $q_\mu^*$ , and  $q_\mu^*(\mathcal{L}(r)) = \mathcal{L}(q_\mu^*(r))$  for restricted  $r$  (item 35).
- Proposition 4.4, the quotient theorem. The paper’s proof is correct modulo the notation swap of item 14; the  $\supseteq$  direction needs  $\otimes^n \mu = \mu$ , which follows from `mu_idem` by induction.
- The equality  $\partial_a^\mu(\mathcal{L}(r)) = \mathcal{L}^\mu(\mathcal{D}_a^\mu(r))$  (item 19).
- **Then the payoff:** fuzzy membership over  $\Sigma$  is crisp membership over  $\Sigma^\mu$ , so the fuzzy DFA is the crisp DFA of Phase 2 applied to  $q_\mu^*(r)$ . That is the cleanest theorem in the paper, it is what line 611 already gestures at, and it is nearly free once Phase 2 exists.

**Needs from the paper:** items 36, 37, and a definition of the fuzzy DFA (F5).

## Phase 8 — Finiteness, and the general case

- Fuzzy finiteness under `mu_idem`, via Phase 7 composed with Phase 2’s finiteness.
- General-norm regularity and finiteness: research, not transcription (items 18 and 20, F3). Budget separately or not at all.

## Ordering, in one paragraph

Phases 0–2 are worth doing whatever you decide about the paper, and Phase 0 in particular should happen this week. Phases 5–7 are the actual contribution, and they are blocked — completely, not partially — on Part B. The realistic shortest path to a paper whose fuzzy half is machine-checked is: Phase 0, Phase 2 (cheap route), Phase 3 with  $\otimes$  instantiated at `min` first so the existing fuzzy files port almost unchanged and act as a regression test for the interface, Phase 5, Phase 7 — and

only then the generalization of Phase 4 to an arbitrary  $\otimes$ . That gives you the quotient theorem and a verified fuzzy automaton for the idempotent case, which is the WiL result stated correctly, and it leaves the general-norm inclusions of Phase 6 as the paper's stated-but-unformalized contribution until the port catches up.

## Part H — Checked and correct: do not spend time here

Everything below I verified while writing this report. It is listed so that nobody re-audits it.

### In the paper

- **Lemma 3.1, both items.** The  $\mu$ -cut is an equivalence when  $\mu \otimes \mu = \mu$  (transitivity via monotonicity), and gödelian  $\Rightarrow$  every  $\mu$ -cut is an equivalence (because  $\otimes$  is min on idempotents). True.
- **Lemma 3.2, all three items.**  $\mathcal{R}^w$ ,  $\mathcal{R}^E$  and  $\mathcal{R}^L$  are T-similarities. I checked reflexivity, symmetry and the reverse triangle inequality clause by clause, including the  $*$  clause and the empty-language cases. Left-continuity is genuinely needed for  $\mathcal{R}^L$  and only for  $\mathcal{R}^L$ .
- **Theorem 3.1(1)**,  $\mathcal{R}^E \leq \mathcal{R}^L \circ \mathcal{L}$ , holds for a general norm. I checked every clause:  $+$  (because  $\mathcal{R}^L(L_1 \cup L_2, M_1 \cup M_2) \geq \min\{\mathcal{R}^L(L_1, M_1), \mathcal{R}^L(L_2, M_2)\}$ ),  $\cdot$ ,  $*$ , letters,  $r = s$ , and the “0 otherwise” case, plus the degenerate  $\emptyset$  cases.
- **Theorem 4.2(3)**, the  $\subseteq$  correspondence, holds for a general norm.
- **Conjecture 3.1 and Lemma 3.3** are both true, and easy (see item 25).
- **Counterexample 1 for Definition 3.4** (the  $*$  one, line 250) is correct.
- **The  $(\neg)$  counterexample’s arithmetic** is correct, including the general formula  $\mathcal{R}^L(\mathcal{L}(\neg a), \mathcal{L}(\neg b)) = \min\{\sup_{c \neq a} \mathcal{R}(a, c), \sup_{c \neq b} \mathcal{R}(b, c)\}$ . Only its framing needs fixing (item 22).
- **Proposition 4.1** is true (given the transfer lemma of item 37), and **Proposition 4.3** is true — and note that the  $\Rightarrow$  direction of Proposition 4.3 needs no hypothesis at all, since  $\otimes \leq \min$ .
- **Proposition 4.4**, the quotient theorem, is true under  $\mu \otimes \mu = \mu$ , and the paper’s proof is correct apart from the notation swap of item 14. So is the displayed consequence, because  $q_\mu^*$  is surjective.
- **Proposition 4.5** (the red one) is true.
- **The extensional-quotient remark** at line 203 is correct, including the implicit claim that  $\mathcal{R}$  is well defined on the classes of its 1-cut.
- **$\mathcal{R}^E$  restricts to  $\mathcal{G}^E$  when  $\otimes = \min$**  — all five clauses agree.
- **Definition 3.5’s overlapping first clause** is harmless: where the clauses overlap, the values agree.
- **Definition A.4**, all four T-norms *and* all four residua, are correct — including the  $p$ -ordinal sum’s three-branch residuum, which I checked case by case. So is the claim at line 205 that these norms are idempotent on  $(0, p]$ .
- **Proposition A.1**, all eight items, are correct — and item (1) is exactly where left-continuity is used, which is worth recording, since the T-norm interface will have to carry it.
- **The Redko / Salomaa / Kozen citations** at line 101 are now correct: Redko for the absence of a finite purely equational axiomatization, Salomaa and Kozen as complete systems that are not purely equational.
- **The LaTeX build is clean:** 0 errors, 0 warnings, 0 undefined references, 0 undefined citations, 0 overfull and 0 underfull boxes.

## Already fixed in the source since the earlier review notes

- Appendix A’s definition of *crisp* now reads  $\mu_S(x) \in \{0, 1\}$  (it used to say “fully included”, which made  $S$  the only crisp set).
- Definition A.4(4) now restricts to  $p \in (0, 1)$ , so the formula no longer divides by zero.
- Definition 4.6 now *has* a base case,  $\partial_\varepsilon^\mu(L) = L^\mu$ . What remains is that  $L^\mu$  is undefined for languages (item 7).
- The empty `\cite{ }` at the old line 313 is gone, and Redko is cited correctly.
- “outside regular languages”  $\rightarrow$  “the”.

## In coq/

- **Languages.v** is well built. `splits` enumerates decompositions into *nonempty* blocks, which is exactly what makes *star* decidable; `conc` and `star` are boolean and executable; `splits_flatten` and the `concP/starP` reflection lemmas are the right interface.
- **Nullable.v** is right, and `nullable r := if has_eps r then Eps else Empty` is the reading the paper should adopt (item 15). `nullable_correct` is proved.
- **Derivatives.v** is the strongest file in the development. `D_char_correct` and `D_correct` are complete, and cover `And` and `Neg` too.
- **Canonicalization.v** implements *exactly* Definition 2.2’s law list — I checked every law against `mkPlus`, `mkConc`, `mkStar`, `mkAnd` and `mkNot`, and each of the three empty-list cases (`Empty`, `Eps`, `Top`) is the correct neutral element. Nothing extra has crept in.
- **CanonicalizationCorrectness.v**: `canonize_correct` is genuinely proved and is the real content. `similar_sound` (paper Lemma 2.2) follows correctly.
- **The empty conventions already match what the paper needs**: `SupL_empty = 0` and `InfL_empty = 1` (item 12).
- `Rw_is_similarity`, `RL_is_similarity`, `Rr_is_similarity` and `syntax_semantics_inequality` are all proved (for `min`). These are paper Lemma 3.2 and Theorem 3.1(1), and they are the part of the fuzzy layer worth porting rather than rewriting.
- **No axioms beyond the alphabet interface**. The only Axioms in the development are the five module-type parameters in **Alphabet.v**, which is what a module type is for. There is exactly one `Admitted` (item 27) and one `TODO` (item 72).

## Checklist before opening Rocq

Tick every line. Each one corresponds to a numbered item above.

### Architecture (Part A)

- ☐ Is the general-norm port of the fuzzy Rocq files scheduled, against a Section 3 that is final? (1)
- ☐ What does the paper claim about the WiL correspondence: inclusion, or equality-under-hypothesis? (2)
- ☐ Are the automata in scope? If yes, the dead block is revived; if no, the abstract changes. (3)
- ☐ Is  $\sim$  equational or `nf`-equality — and is it stated to be a congruence? (4)
- ☐ Which equivalence does the automaton use, now that  $\mathcal{D}$  is not a congruence for  $\sim$ ? (5)

### Well-formedness (Part B) — the hard gate

- ☐ Every fuzzy definition is a finite, well-founded, single-valued construction. (6, 11, 13)
- ☐  $\mathcal{L}^\mu$  and  $\sim_{\mu}^{\Sigma^*}$  are defined in Section 3, for languages, unconditionally. (7)
- ☐ Every theorem is stated over the domain its ingredients are defined on. (8, 10)
- ☐ The empty-product,  $\inf \emptyset$ ,  $\sup \emptyset$  and  $\otimes^n$  conventions are written down. (12)
- ☐  $[\cdot]$  and  $\mathcal{L}$  each mean one thing. (14)
- ☐ The restricted fragment is genuinely closed under everything applied to it. (15)

### Truth (Part C)

- ☐ No theorem silently relies on  $\mu \otimes \mu = \mu$ ; each that needs it says so. (18–21)
- ☐ No statement in the paper contradicts another. (2, 19, 21, 24)
- ☐ Every false statement is either repaired or deleted. (21–24, 28, 29, 31–34)
- ☐ The two missing lemmas are stated. (36, 37)
- ☐ Every direction that is not known to hold is flagged as such, so that nobody sets out to formalize it blind. (18, 20, 27)

### Alignment with the code (Part E)

- ☐ `make` completes. (64)
- ☐ One `regex` type. (65)
- ☐ One restricted-fragment notion, used where the paper says “restricted”. (68)
- ☐ `README.md` lists every numbered result in `dhkm.tex` with an honest status. (76)
- ☐ The abstract’s claim about what is formalized is true. (59)

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Every counterexample in this report was computed from the paper’s own definitions and, where the definitions exist in `coq/src/`, against the code’s own `canonicalize`. Where I state that something is true, I checked it; where I could not settle it, the item says **[UNPROVED]** rather than **[FALSE]**. The three statements I could neither establish nor

refute are Theorem 3.1(2)'s  $\geq$  direction (item 27), general-norm regularity (item 18), and general-norm finiteness (item 20) — those three are the real mathematical risk in the formalization, and nothing else on this list is.